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The Plateau Problem and Minimal Surface Theory: An Exposition of Douglas's Proof

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Abstract

In this thesis, we give an introduction to Plateau's Problem and the theory of minimal surfaces. Given a simple closed curve Γ, Plateau's Problem asks whether there exists a surface of minimal area whose boundary is Γ. Minimal surfaces naturally arise in the study of area-minimizing surfaces, as they include such surfaces and are characterized by having zero mean curvature, $H = 0$.

The first half of the thesis is dedicated to studying the properties of minimal surfaces, with a focus on their connections to various branches of mathematics and the many equivalent ways in which they can be defined. Historically, the question of area-minimizing surfaces has been notoriously difficult to tackle. In the second half, we present Douglas's theorem (1931), along with a modernized proof showing that an area-minimizing surface always exists for a Jordan curve in $\mathbb{R}^n$. We explore the key ideas behind the proof, emphasizing the role of isothermal parametrizations and describing the conditions under which they exist. We end by comparing the present version with Douglas's original argument and discussing the more recent directions in the study of Plateau's Problem.

Notation

Topology and metric spaces

For a metric space (X, d), $x \in X$, $U \subseteq X$, we denote:

- *Interior and Closure*: $\mathrm{int}(U) = \bigcup_{O \subseteq U, \text{ open}} O, \quad \overline{U} = \bigcap_{U \subseteq C, \text{ closed}} C$
- *Boundary*: $\partial U = \overline{U} \setminus \mathrm{int}(U)$
- *Open and Closed balls*: $B(x, r) := \{y \in X \mid d(x, y) < r\}, \quad \overline{B}(x, r) := \{y \in X \mid d(x, y) \leq r\}$

Multivariate calculus in Euclidean space

We use $\mathbb{R}^n$ to denote n-dimensional Euclidean space with the standard topology and $\mathcal{B}_c = (e_1, \ldots, e_n)$ the canonical basis. For $x, y \in \mathbb{R}^n$ and $r > 0$, we denote:

- *Euclidean inner product*: $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$
- *Euclidean norm*: $\|x\| = \sqrt{\langle x, x \rangle}$
- *Euclidean distance*: $d(x, y) = \|x - y\|$
- *Angle between two vectors*: $\cos\theta = \frac{\langle x, y \rangle}{\|x\|\|y\|}$
- If $n = 3$, *Cross product*: $x \times y = (x_2 y_3 - x_3 y_2,\ x_3 y_1 - x_1 y_3,\ x_1 y_2 - x_2 y_1)$
- *n-sphere*: $\mathbb{S}^n = \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1^2 + \ldots x_{n+1}^2 = 1\}$

Let $U \subseteq \mathbb{R}^n$, $p \in U$, $f : U \to \mathbb{R}^m$ a function with components f_j, we denote:

- *Partial derivative*: $\left.\frac{\partial f}{\partial x_j}\right|_p = \partial_j f|_p = f_{x_j}(p) = \lim_{t \to 0} \frac{f(p + t e_j) - f(p)}{t}$
- If f is $k \in \mathbb{N} \cup \{\infty\}$ times continuously differentiable with α-Hölder continuous k-th derivative then $f \in \mathcal{C}^{k,\alpha}(U)$
- *Differential Map*: $df : \mathbb{R}^n \to \mathbb{R}^m$ and *Jacobian matrix*: $Jf(p) := [df]_{\mathcal{B}_c}(p)$
- If $m = 1$, *Gradient*: $\nabla f = Df = (\partial_1 f, \ldots, \partial_n f)$
- If $n = m$, *Divergence*: $\mathrm{div}(f) = \sum_{j=1}^n \partial_j f_j$
- If $f \in \mathcal{C}^2(U)$, *Laplacian*: $\Delta f = \mathrm{div}(\nabla f) = \sum_{j=1}^n \partial_{jj} f$
- If $n = m$ and $u_j := f_j$, *Jacobian determinant*: $\frac{\partial(u_1, \ldots, u_n)}{\partial(x_1, \ldots, x_n)} = \det(Jf)$
- *Compactly supported infinitely differentiable functions*: $f \in \mathcal{C}_c^\infty(U)$

Complex analysis

We denote with $\mathbb{C}$ the set of complex numbers. For $U \subseteq \mathbb{C}$, a complex function $f : U \to \mathbb{C}$ and $z = x + iy \in U$, we denote:

- *Conjugate*: $\overline{z} = x - iy$
- *Modulus*: $|z| = \sqrt{z\overline{z}} = \sqrt{x^2 + y^2}$
- *Complex derivative*: $f'(z) = \lim_{w \to z} \frac{f(w) - f(z)}{w - z}$ and *k-th derivative*: $f^{(k)}(z)$
- *Wirtinger derivatives*: $\partial_z f = \frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right), \quad \partial_{\overline{z}} f = \frac{\partial f}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right)$
- *Open unit disk*: $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$
- *Extended complex plane*: $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$

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0 Introduction

To motivate the study of Plateau's Problem it is helpful to first look at a simpler problem. Take two points A and B in the Euclidean plane, a natural question one may ask is *what is the shortest path between them?* It is a well-known fact that the answer is the straight line joining A and B. One can take this question a step further by considering that A and B lie on a connected two-dimensional surface S embedded in three dimensional space; in this case, the answer is no longer obvious. The length of a smooth curve $\gamma : [a, b] \to S$ is given by

$$\ell(\gamma([a, b])) = \int_a^b \|\gamma'(s)\| \, \mathrm{d}s.$$

Thus, we may formally state our previous question as asking for a curve γ satisfying $\gamma(a) = A$ and $\gamma(b) = B$ such that $\ell(\gamma([a, b]))$ is minimized among all such curves. At first, it is not even clear whether such a curve exists, let alone what it might look like for specific points A and B on a given surface S. As it turns out, under fairly general assumptions on the surface S, such length-minimizing curves do exist and are known as *geodesics*. More precisely, geodesics form a broader class of curves that can be defined in several equivalent ways, one of which is as critical points of the length functional. That is, a curve γ is a geodesic if

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \int_a^b \big\|(\gamma + t\eta)'(s)\big\| \, \mathrm{d}s = 0$$

for any arbitrary smooth variation $\eta : [a, b] \to S$ vanishing at the endpoints, i.e. $\eta(a) = \eta(b) = 0$, see Fig. 0.1. It is a classical result that any curve minimizing length must be a geodesic.

Geodesics play a central role in many areas of mathematics and physics. For instance, in General Relativity, they describe the motion of free-falling particles. Since curves are one-dimensional objects whose boundaries consist of zero-dimensional points, a natural follow-up question is to look for a two-dimensional surface bounded by a fixed one-dimensional curve that minimizes area among all such surfaces. The question of existence of such an area-minimizing surface is classically known as *Plateau's Problem*. *Minimal surfaces* are the two-dimensional analogues of geodesics, defined as the critical points of the area functional.

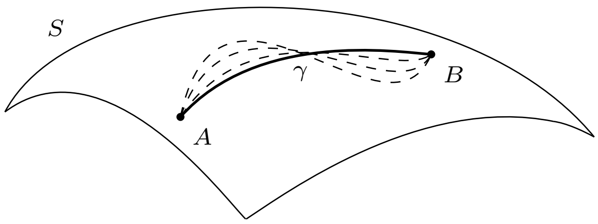


Figure 0.1: A curve γ minimizing the distance between the points A and B on a surface S and its variations.

Plateau's Problem was first posed and studied by Lagrange in 1760, who approached it using his newly developed tools from the calculus of variations. Since then, the theory of minimal surfaces has evolved significantly. As described by Joaquín Pérez in his article «A new golden age of minimal surfaces» [Pér16], the development of minimal surface theory can be divided into distinct "ages", each marked by major mathematical advances:

- *The First Golden Age of Minimal Surfaces* (ca. 1830–1890): Nearly a century after Lagrange, in 1849, Joseph Plateau approached the problem from a physical perspective by observing soap films formed on wire frames. These films naturally form surfaces that minimize surface tension, and hence area, providing physical evidence of the existence of area-minimizing surfaces. Plateau's emphasis on the physical problem earned him the honor of having the problem named after him. Around the same time, mathematicians such as Enneper, Schwarz, Riemann, and Weierstrass made foundational contributions to the theory of minimal surfaces. In Chapter 1, we examine the origins of Plateau's Problem and the main mathematical advances of this period.

- *The Second Golden Age of Minimal Surfaces* (ca. 1914–1950): In 1931, the mathematicians Jesse Douglas and Tibor Radó proved independently the existence of an area minimizing surface having the topology of the disk. Particularly, the method used by Douglas in his proof, known as the *direct method*, has proved to have extensive applications in similar problems. Because of his work on Plateau's Problem, Douglas was awarded one of the first two Fields Medals in 1936. This period also saw contributions from other major figures such as Riemann, Bernstein, and Courant, marking a second highly productive era in the development of minimal surface theory. In Chapter 2, we present a modernized version of Douglas's proof, highlighting its central ideas and comparing them to his original argument.

- *The Third Golden Age of Minimal Surfaces* (ca. 1960–1980): Starting in the 1960s, a new wave of results on minimal surfaces began to appear, thanks to the advent of new techniques from Riemannian geometry, the theory of Riemann surfaces, geometric measure theory, functional analysis, and more. These came from the work of major figures like Calabi, do Carmo, Federer, Fleming, and Osserman. In Chapter 3, we informally go over some of these ideas and what motivated their introduction.

In this thesis, we develop the theory of minimal surfaces with a focus on their relationship to Plateau's Problem. We will see that it is a rich theory, connecting topics in differential geometry, the theory of partial differential equations and harmonic functions, complex and functional analysis, and the calculus of variations. The structure of this work aims to mimic the historical development of Plateau's Problem, beginning with its initial formulation and how it evolved into a central topic within the broader theory of minimal surfaces. We then examine in depth how the problem was first solved and how the proof has evolved over the course of nearly a century. Finally, we quickly discuss the recent advances in the theory, going over alternative formulations of Plateau's problem in higher dimensions and the modern methods developed to tackle it. In summary, the goal of this thesis is to give a thorough introduction to Plateau's Problem, allowing any mathematically inclined person that has never heard of minimal surfaces to get a sense of how beautiful the theory of minimal surfaces can be and how technical the proofs can get.

Since the theory of minimal surfaces touches on many areas of mathematics, we assume the reader is comfortable with basic multi-variable calculus, topology, and both real and complex analysis. Some background in differential geometry (particularly the theory of surfaces) and in PDEs and harmonic functions is also helpful. Because complex analysis and harmonic functions are particularly relevant, a brief review is provided in Appendix A, covering all the results used throughout the thesis. Finally, although the thesis has been written avoiding the technicalities of the Lebesgue integral, it is true that every instance in which an integral is involved can be more cleanly treated and generalized using the tools of measure theory. With the idea of a self-contained thesis in mind, Appendix B includes the Riemann integral version of the three main integration results which will be used extensively: the integration by parts formula, differentiation under the integral sign, and the fundamental lemma of the calculus of variations.

1 Introduction to Plateau's Problem and minimal surfaces

In this first chapter, we examine the initial developments of Plateau's Problem and how it naturally gave rise to the theory of minimal surfaces. We begin by formulating the problem in mathematically precise terms. We then study minimal surfaces in $\mathbb{R}^3$, focusing on their various characterizations: as geometric objects with vanishing mean curvature, as surfaces defined by real harmonic functions, and as images of holomorphic maps in complex analysis. In the study of minimal surfaces via complex functions, we derive what is arguably the most important result of the first golden age of minimal surfaces: the Weierstrass–Enneper data. Finally, we study the catenoid, the first non-trivial example of a minimal surface, whose properties provide insight into the broader theory and Plateau's Problem.

1.1 The original problem

What keeps Plateau's problem relevant to this day is the fact that it can be posed in many ways with increasing complexity each of which may be tackled more appropriately by a certain set of mathematical tools than others. In order to have a well posed version of Plateau's problem it is necessary to first define the notions of curve, surface and area which ideally we will be able to intuitively connect to some physical representation of the problem. As Lagrange first did, we will begin by considering surfaces as the graphs of functions.

Definition 1.1.1. Let $f : U \to \mathbb{R}$ where $U \subseteq \mathbb{R}^2$ is open, we define the surface determined by the *graph of* f as the set

$$G_f := \{(x, y, z) \in \mathbb{R}^3 \mid (x, y) \in U,\ z = f(x, y)\} \subseteq \mathbb{R}^3.$$

Such surfaces are also known historically as *non-parametric* surfaces. We can now define the area of G_f according to basic multivariate calculus.

Definition 1.1.2. Let $f \in \mathcal{C}^1(U)$, we define the *area of the graph* of f by

$$A(f) := \int_U \sqrt{1 + \|\nabla f\|^2}\, \mathrm{d}x\mathrm{d}y = \int_U \sqrt{1 + f_x^2 + f_y^2}\, \mathrm{d}x\mathrm{d}y.$$

Remark 1.1.3. For the area of G_f to be well defined we require f to be at least $\mathcal{C}^1$ but for purposes that we will see shortly we will sometimes be more interested in requiring f being $\mathcal{C}^2$.

We remind ourselves that the previous definition can be justified geometrically by considering the tangent vectors of G_f at a given point p which can be obtained by differentiating $F(x, y) = (x, y, f(x, y))$ in each direction. Then at an «infinitesimal» level the area $\mathrm{d}A$ will be that of the parallelogram spanned by these two vectors such that

$$\mathrm{d}A = \left\|F_x \times F_y\right\| \mathrm{d}x\mathrm{d}y = \left\|(1, 0, f_x) \times (0, 1, f_y)\right\| \mathrm{d}x\mathrm{d}y = \sqrt{1 + f_x^2 + f_y^2}\, \mathrm{d}x\mathrm{d}y.$$

We now suppose that $\Gamma \subset \mathbb{R}^3$ is a simple closed curve with finite length that is the graph of some continuous function $\gamma : \partial U \to \mathbb{R}$. One of the first formulations of Plateau's problem is the following:

Let $U \subseteq \mathbb{R}^2$ be a simply connected open set and fix $\gamma \in \mathcal{C}^0(\partial U)$ such that $\Gamma := G_\gamma$ is a continuous simple closed curve. Is there $f \in \mathcal{C}^1(U) \cap \mathcal{C}(\overline{U})$ such that $A(f)$ is minimal among all f having $f|_{\partial U} = \gamma$?

In this formulation, the problem can be easily restated in a form which is much more approachable from a computational point of view by using the Calculus of Variations. The idea consists in taking some arbitrary smooth function that vanishes on the boundary $\eta \in \mathcal{C}_c^\infty(\overline{U})$ meaning that if $f : \overline{U} \to \mathbb{R}$ is some function having $f|_{\partial U} = \gamma$ then $f + t\eta$ will also agree with γ on ∂U for any $t \in (-\epsilon, \epsilon)$. We say that the graph of $f + t\eta$ is a *perturbation of the graph* of f. Now if f is a function that minimizes area among small perturbations then $A(f) \leq A(f + t\eta)$ for small enough t and any η implying

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} A(f + t\eta) = 0.$$

Exchanging the derivative with the integral we find that

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} A(f + t\eta) = \int_U \frac{\partial}{\partial t}\Big|_{t=0} \sqrt{1 + \|\nabla(f + t\eta)\|^2}\, \mathrm{d}x\mathrm{d}y = \int_U \frac{\langle \nabla f, \nabla \eta \rangle}{\sqrt{1 + \|\nabla f\|^2}}\, \mathrm{d}x\mathrm{d}y$$

and if f is $\mathcal{C}^2$ and ∂U regular enough we can integrate by parts, with a vanishing boundary term since $\eta|_{\partial U} \equiv 0$, to find that

$$\int_U \operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + \|\nabla f\|^2}} \right) \eta \, \mathrm{d}x\mathrm{d}y = 0$$

must hold for all η. This implies that the term involving the divergence must vanish yielding the following partial differential equation

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1 + \|\nabla f\|^2}} \right) = \frac{(1 + f_y^2) f_{xx} - 2 f_x f_y f_{xy} + (1 + f_x^2) f_{yy}}{(1 + \|\nabla f\|^2)^{3/2}} = 0,$$

or equivalently

$$(1 + f_y^2) f_{xx} - 2 f_x f_y f_{xy} + (1 + f_x^2) f_{yy} = 0 \tag{1.1.1}$$

which is know as the *equation for minimal graphs* of dimension 2.

Definition 1.1.4. A solution to Eq. (1.1.1) is said to be a *minimal graph surface*.

Equation (1.1.1) is a second order partial differential equation which is quasi-linear, meaning that is linear in the highest order derivatives. As a matter of fact, by considering very small first order derivatives one can approximate the previous PDE to *Laplace's equation*, that is if $f_x, f_y \simeq 0$ then the equation becomes

$$f_{xx} + f_{yy} = 0.$$

The connection to Laplace's equation will be recurring during the study of minimal surfaces from the classical point of view.

Remark 1.1.5. It is enough to consider variations of f of the form $f + t\eta$ since a more general variation $f^t(x, y) = F(x, y, t)$ that verifies $F(x, y, 0) = f(x, y)$ and $F(x, y, t) = f(x, y)$ for $(x, y) \in \partial U$ will only affect the area up to first order. Indeed, by Taylor expanding F we get

$$F(x, y, t) = f(x, y) + t\partial_t F(x, y, 0) + O(t^2)$$

so terms of order t^2 or higher will vanish when differentiating and setting $t = 0$. Renaming $\eta(x, y) := \partial_t F(x, y, 0)$ we see that it is enough to consider variations of the previous form.

A comment needs to be with respect to the relationship between minimal graphs and solutions to Plateau's Problem. It is clear that if the graph of a certain $\mathcal{C}^2$ function f is absolutely minimizing of the area functional in the class of all $\mathcal{C}^1$ functions with the same bounding curve then it will be a solution to the equation for minimal graphs, yet the converse is not necessarily true. This is because the procedure used to derive Eq. (1.1.1) only allows us to detect locally minimizing graphs for the same reasons that finding the zeroes of a functions derivative only allows us to find local extrema. Moreover, there might be a $\mathcal{C}^1$ function that minimizes $A(f)$ yet is not twice differentiable such that it makes no sense to check if it satisfies Eq. (1.1.1) (in the classical sense).

Remark 1.1.6. In higher dimensions minimal graph hypersurfaces are defined by $f : U(\subseteq \mathbb{R}^n) \to \mathbb{R}$ satisfying

$$\operatorname{div}\left(\frac{\nabla f}{\sqrt{1+\|\nabla f\|^2}}\right) = 0.$$

This equation trivially has linear functions $a_1x_1 + \dots a_nx_n + c$ as solutions defined over the entire space $U = \mathbb{R}^n$. The question asking if a minimal hypersurface that is a graph of an entire function $f : \mathbb{R}^n \to \mathbb{R}$ for $n \geq 2$ has to be planar is known as *Bernstein's problem* and the answer is affirmative for $n \leq 7$. For $n = 8$ Bombiere, De Giorgi & Giusti found that there exist non-planar entire minimal surfaces [BGG69]. This is in stark contrast with the Laplace equation which admits many global solutions in any dimension.

We notice that by restricting our attention to surfaces that are graphs of functions we are dismissing a large amount of very interesting and physically relevant cases such as the surface minimizing the area bounded by the two bounding circles of a given cylinder. This particular example is very well suited to real-world experimentation by creating soap bubble films between two circular wires as Plateau did in his experiments. We might also be interested in minimizing surfaces bounded by curves that live in higher dimensions. Motivated by such cases we aim to extend our definition of a surface and its area.

Definition 1.1.7. A differentiable map $f : D \subseteq (\mathbb{R}^n) \to \mathbb{R}^m$ with $n \leq m$ is known as an *immersion* if its differential map df is injective everywhere.

Immersions allows us to extend our notion of surfaces.

Definition 1.1.8. We say $S \subseteq \mathbb{R}^n$ is a $\mathcal{C}^k$ *immersed surface* if it is locally the image of a $\mathcal{C}^k$ immersion: for any $p \in S$ there exists an open set $D \subseteq \mathbb{R}^2$ and a $\mathcal{C}^k$ immersion $\varphi : D \to \mathbb{R}^n$ such that $p \in \varphi(D) \subset S$. The pair (D, φ) is known as a *local parametrization* of S.

Moreover, a point $p \in S$ of an immersed surface is *regular* if there exists an open $V \subseteq \mathbb{R}^n$ containing p and an open $U \subset D$ such that $\varphi : U \to S \cap V$ is a homeomorphism. If all points of S are regular then we say that S is a *regular surface.*

In general, immersed surfaces can exhibit self-intersections and other pathologies, whereas regular surfaces align more closely with our intuitive idea of what a surface should be. At first glance, then, it might seem more appropriate to work only with regular surfaces. However, by restricting the domain of a parametrization of an immersed surface, we can always obtain a regular surface. Moreover, any regular surface is locally the graph of a function. These are consequences of the inverse and implicit function theorems, respectively.

Proposition 1.1.9.

(a) If $S \subset \mathbb{R}^n$ is an immersed surface parametrized by (D, φ) then for every $p \in S$ there exists $D' \subseteq D$ such that $\varphi(D')$ is a regular surface containing p.

(b) If $S \subset \mathbb{R}^n$ is a regular surface then for every $p \in S$ there exists an open $V \subseteq \mathbb{R}^n$ containing p such that $S \cap V$ is the graph of a function.

We will usually refer to both immersed and regular surfaces as just surfaces and make the distinction only when regularity is necessary. We will denote $(u, v) \in D$ the *parameters* and $(x^1, \dots, x^n) = \varphi(u, v) \in S$ the *coordinates* of a surface. We can now extend our definition of area to accommodate this broader class of surfaces.

Definition 1.1.10. We define the *area of a $\mathcal{C}^1$ immersed surface $S = \varphi(D)$* via the following formula

$$A(S) := \int_D \sqrt{\|\varphi_u\|^2 \|\varphi_v\|^2 - \langle \varphi_u, \varphi_v \rangle^2} \, \mathrm{d}u\mathrm{d}v \,. \tag{1.1.2}$$

The previous definition can easily be verified to be independent of the choice of parametrization with the change of variables formula. This justifies the lack of reference to the parametrization in the notation $A(S)$ and implies that the area is a geometrical property of S.

Remark 1.1.11. A surface that is a graph $S = G_f$ is a particular case of a regular surface in which we take the parametrization $\varphi(u, v) = (u, v, f(u, v))$. If $n = 3$ equation (1.1.2) can also be written as

$$A(S) := \int_D \|\varphi_u \times \varphi_v\| \; \mathrm{d}u\mathrm{d}v,$$

which allows us to give the same geometrical meaning of the integrand as in the case of graphs.

By considering immersed surfaces that are bounded curves in $\Gamma \subseteq \mathbb{R}^n$ homeomorphic to the circle $\mathbb{S}^1$, we can restate Plateau's Problem in a more general form as follows:

Let $\Gamma \subseteq \mathbb{R}^n$ be a curve homeomorphic to the circle. Is there $\gamma : \mathbb{S}^1 \to \Gamma$ a parametrization of Γ and an immersion $\varphi : \overline{\mathbb{D}} \to \mathbb{R}^n$ which is $\mathcal{C}^1$ in it's interior and continuous up to its boundary such that $\varphi|_{\partial\mathbb{D}} = \gamma$ and $S = \varphi(D)$ has the least area among all such surfaces?

This version of Plateau's Problem is (almost) precisely the one solved independently by Douglas and Radó. We note that we have slightly restricted our definition of an immersed surface by requiring that $D = \mathbb{D}$. This simplification not only makes the problem more manageable but is also necessary, since allowing more general domains permits solutions of varying topological type. We return to this point for further discussion in §2.1.

Remark 1.1.12. The question asking if an area minimizing surface of a given Plateau Problem is a regular surface or not is another within the field of minimal surfaces. An important related result is that of Meeks and Yau [MY82] which states that if Γ lies in the boundary of a convex subset of $\mathbb{R}^3$ then the solution to Plateau's Problem is embedded.

1.2 The geometry of minimal surfaces in $\mathbb{R}^3$

Until the end of this chapter we will focus our attention on surfaces in $\mathbb{R}^3$. Our aim now is to derive an analog of the equation for minimal graphs for parametrized surfaces, this will lead us naturally to one of the key properties of area minimizing surfaces:

Surfaces that minimize area have vanishing mean curvature everywhere.

To make sense of this statement we will start by looking at the case of regular surfaces. After reviewing the fundamental concepts of the theory of smooth regular surfaces in 3 dimensions (a more detailed read on the theory can be found in [Pre10]) we will be able to define them for immersed surfaces which will allow us to derive an equation for parametric minimal surfaces.

Definition 1.2.1. The *tangent space* of a regular surface $S \subseteq \mathbb{R}^3$ at a point $p \in S$ is defined as

$$T_pS := \operatorname{Im} d\varphi_{\varphi^{-1}(p)}$$

where (D, φ) is a local parametrization around p and $d\varphi$ is the differential map of φ.

From the definition T_pS will be a 2 dimensional sub-vector space of $\mathbb{R}^3$ and it is that spanned by φ_u and φ_v. It can also be seen that T_pS is the set of vectors $X \in \mathbb{R}^3$ such that there exists a smooth curve $\gamma : (-\epsilon, \epsilon) \to S$ having $\gamma(0) = p$ and $\gamma'(0) = X$. So the tangent space coincides with our geometrical intuition of what a surface's tangent plane at a point should be and it is a geometrical invariant, i.e., it does not depend on φ. Now, since T_pS is a finite dimensional real vector space it can be endowed with an inner product and the natural choice is the following.

Definition 1.2.2. The positive definite, symmetric and bilinear map

$$\begin{aligned} \mathrm{I}_p : T_pS \times T_pS &\to \mathbb{R} \\ (X, Y) &\mapsto \langle X, Y \rangle \end{aligned}$$

is called the *first fundamental form* of S at p. If (D, φ) is a parametrization of S then we define

$$E = \langle \varphi_u, \varphi_u \rangle, \quad F = \langle \varphi_u, \varphi_v \rangle, \quad G = \langle \varphi_v, \varphi_v \rangle \tag{1.2.1}$$

the *local components* of I_p in (D, φ).

Remark 1.2.3. The fact that φ is an immersion is equivalent to saying that $EG - F^2 > 0$. We also note that the integrand in the definition of the area for an immersed surface, Eq. (1.1.2), can be re-written as $\sqrt{EG - F^2} = \sqrt{\det[\mathrm{I}_p]_{\mathcal{B}}}$ where $[\,]_{\mathcal{B}}$ denotes the matrix representation in the basis $\mathcal{B} := (\varphi_u, \varphi_v)$.

Given that T_pS is a 2 dimensional subvector space of $\mathbb{R}^3$, for each $p \in S$ there is a well defined orthogonal 1 dimensional vector space $(T_pS)^\perp$.

Definition 1.2.4. We say that S is *orientable* if there exists a smooth map $\nu : S \to \mathbb{S}^2$ such that $T_pS = \nu(p)^\perp$. Such a map ν is known as the *Gauss map* or a *normal unit vector field* of S.

If S is orientable with Gauss map ν then it also admits $-\nu$ as a Gauss map and this is the only other possibility. A choice of sign between ν and $-\nu$ defines an *orientation* of S. We notice that if S is orientable and φ a parametrization then the map

$$\nu_\varphi(p) := \frac{\varphi_u \times \varphi_v}{\|\varphi_u \times \varphi_v\|} \circ \varphi^{-1}(p) \tag{1.2.2}$$

is a Gauss map and defines an orientation of S.

From now on we will suppose that $\varphi \in \mathcal{C}^2(D)$ such that it induces a differentiable Gauss map ν_φ. We note that $T_pS = \nu(p)^\perp = T_{\nu(p)}\mathbb{S}^2$ and this fact allows us to make the following definition.

Definition 1.2.5. The linear map

$$\begin{aligned} W_p : T_pS &\to T_pS \, (= T_{\nu(p)}\mathbb{S}^2) \\ X &\mapsto -d\nu_p(X) \end{aligned}$$

is known as the *Weingarten endomorphism* or *shape operator* of S at p.

From the fact that $d\nu_p(X) = \partial_X \nu(p)$ where ∂_X is the directional derivative in the direction $X \in \mathbb{R}^3$ we see that Weingarten endomorphism measures the rate of change of $\nu(p)$ in each direction.

Definition 1.2.6. The bilinear map

$$\begin{aligned} \mathrm{II}_p : T_pS \times T_pS &\to \mathbb{R} \\ (X, Y) &\mapsto \mathrm{I}_p(W_p(X), Y) =: \mathrm{II}_p(X, Y) \end{aligned}$$

is known as the *second fundamental form* of S at p.

Unlike the first fundamental form, II_p is not positive definite but as the following result shows it is still symmetric.

Lemma 1.2.7. *The Weingarten endomorphism is self-adjoint with respect to the first fundamental form*

$$\langle W_p(X), Y\rangle = \langle X, W_p(Y)\rangle.$$

Proof. See Lemma 7.2.3 and Corollary 7.2.4 of [Pre10]. □

Because of this result we also know by the spectral theorem that W_p must have real eigenvalues corresponding to eigenvectors which constitute an orthonormal basis of T_pS.

Definition 1.2.8. The eigenvalues $k_1(p), k_2(p)$ of the Weingarten endomorphism are known as the *principal curvatures* of S at p. We also define the quantities

$$K(p) := \det W_p = k_1(p)k_2(p), \quad H(p) := \frac{1}{2}\operatorname{Tr} W_p = \frac{k_1(p)+k_2(p)}{2}, \tag{1.2.3}$$

which are known as the *Gauss curvature* and *mean curvature*, respectively.

Remark 1.2.9. In some literature the mean curvature is defined as just the trace of W_p, in practice this only affects computations by a factor.

All the quantities defined above are geometrical properties of S, meaning that they are independent of the choice of parametrization. However, it is important to distinguish between intrinsic and extrinsic properties: an intrinsic property can be computed from measurements within the surface itself while an extrinsic property depends on how the surface is embedded in the ambient space. In particular, the Gaussian curvature is an intrinsic quantity, while the principal curvatures and the mean curvature are extrinsic. Moreover, the principal and mean curvatures are dependent on the choice of orientation and choosing $-\nu$ instead of ν changes their sign. It is because of this that is convenient to work with the following vector.

Definition 1.2.10. The vector $\vec{H}(p) := H(p)\nu(p)$ is independent of the choice of orientation and is known as the *mean curvature vector*.

As we have mentioned, if S is orientable then a parametrization (D,φ) induces an orientation ν_φ given by Eq. (1.2.2) from which we can compute local coordinate expressions for W_p and II_p.

Proposition 1.2.11. *Let $S \subseteq \mathbb{R}^3$ be an orientable regular surface oriented by $\nu_\varphi(p)$ where (D,φ) is a local parametrization. Let $\hat{\nu}_\varphi := \nu_\varphi \circ \varphi$ and define*

$$L := \langle \hat{\nu}_\varphi, \varphi_{uu}\rangle, \quad M := \langle \hat{\nu}_\varphi, \varphi_{uv}\rangle, \quad N := \langle \hat{\nu}_\varphi, \varphi_{vv}\rangle,$$

then in the basis $\mathcal{B} := (\varphi_u, \varphi_v)$ of T_pS we have

$$[\mathrm{II}_p]_{\mathcal{B}} = \begin{pmatrix} L & M \\ M & N \end{pmatrix} \quad \textit{and} \quad [W_p]_{\mathcal{B}} = \frac{1}{EG-F^2}\begin{pmatrix} LG-MF & MG-NF \\ -LF+ME & -MF+NE \end{pmatrix}.$$

Proof. See Proposition 8.1.2 of [Pre10]. □

An immediate consequence of the previous formula for W_p is the following local expressions for the Gauss and mean curvatures

$$K = \frac{LN-M^2}{EG-F^2}, \quad H = \frac{LG-2MF+NE}{2(EG-F^2)}, \tag{1.2.4}$$

and these expressions allow us to define the Gauss and mean curvatures for immersed surfaces.

Definition 1.2.12. Let $S = \varphi(D)$ be an immersed surface with $\varphi \in \mathcal{C}^2(D)$. We define the *parametric Gauss and mean curvature* of S according to the expressions in Eq. (1.2.4) where

$$E = \langle \varphi_u, \varphi_u \rangle, \quad F = \langle \varphi_u, \varphi_v \rangle, \quad G = \langle \varphi_v, \varphi_v \rangle$$

are the *parametric first fundamental form coefficients* and

$$L := \langle \hat{\nu}, \varphi_{uu} \rangle, \quad M := \langle \hat{\nu}, \varphi_{uv} \rangle, \quad N := \langle \hat{\nu}, \varphi_{vv} \rangle;$$

are the *parametric second fundamental form coefficients* with $\hat{\nu} := \varphi_u \times \varphi_v / \|\varphi_u \times \varphi_v\|$ the *parametric Gauss map.*

Remark 1.2.13. At regular points of S these quantities will coincide with those of the corresponding regular surface given by Prop. 1.1.9. However at non-regular points such as self-intersections K and H might take on multiple values, one for each sheet of the surface passing through p.

We now proceed to study the connection between curvature and area of an immersed surface. As we did for graphs, given an immersed surface S with parametrization (D, φ) we will consider a small perturbation S^t that fixes the boundary given by

$$\varphi^t(u, v) = \varphi(u, v) + t\eta(u, v)\hat{\nu}(u, v) \tag{1.2.5}$$

where $\eta \in \mathcal{C}_c^\infty(\overline{D})$. We note that variations of this form are not the most general kind, as we only take the variation in the normal direction which is why they are known as *normal variations.* It can be shown that variations in the tangential directions of S do not modify the area (for a proof see [Pre10] chapter 12) so that considering only normal ones is sufficient. The following lemma shows how normal variations impact the first fundamental form up to first order.

Lemma 1.2.14. *Let S be an immersed surface and suppose S^t is a normal variation of the form of Eq.* (1.2.5) *with first form coefficients E^t, F^t and G^t. If ϵ is small enough then S^t will be an immersed surface for each $t \in (-\epsilon, \epsilon)$ and*

$$\frac{\partial}{\partial t}\Big|_{t=0} E^t G^t - (F^t)^2 = -4\eta H(EG - F^2).$$

Proof. We will have that

$$\varphi_u^t = \varphi_u + t\left(\hat{\nu}_u \eta + \hat{\nu}\eta_u\right), \quad \varphi_v^t = \varphi_v + t\left(\hat{\nu}_v \eta + \hat{\nu}\eta_v\right),$$

so we can compute the coefficient

$$E^t = \langle \varphi_u^t, \varphi_u^t \rangle = E + 2t\eta\langle \varphi_u, \hat{\nu}_u \rangle + O(t^2)$$

where we have used the fact that $\langle \hat{\nu}, \varphi_u \rangle = 0$. Differentiating the previous fact with respect to u we also we find that $\langle \varphi_u, \hat{\nu}_u \rangle = -\langle \varphi_{uu}, \hat{\nu} \rangle = -L$ and we get that

$$E^t = E - 2t\eta L + O(t^2);$$

with an analogous procedure we find the remaining coefficients

$$F^t = F - 2t\eta M + O(t^2), \quad G^t = G - 2t\eta N + O(t^2).$$

We will then have that

$$E^tG^t - (F^t)^2 = EG - F^2 - 2t\eta(LG + NE - 2MF) + O(t^2)$$

which will be positive for t small enough, implying S^t is immersed. Finally,

$$\frac{\partial}{\partial t}\Big|_{t=0} E^tG^t - (F^t)^2 = -2\eta(LG + NE - 2MF) = -4\eta H(EG - F^2). \qquad \square$$

Like in the case of graphs, if S is an area minimizing surface amongst small normal variations then $A(S) \leq A(S^t)$ implying that

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} A(S^t) = 0,$$

and the previous result allows us to compute the derivative of $A(S^t)$.

Theorem 1.2.15 (First Variation of Area)**.** *Let S^t be a normal variation of the form of Eq.* (1.2.5)*, then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} A(S_t) = -2\int_D \eta H\sqrt{EG - F^2}\,\mathrm{d}u\mathrm{d}v.$$

Proof. Exchanging the derivative with the integral and applying the previous lemma we find that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} A(S_t) &= \int_D \frac{\partial}{\partial t}\Big|_{t=0} \sqrt{E^tG^t - (F^t)^2}\,\mathrm{d}u\mathrm{d}v \\ &= \int_D \frac{\partial}{\partial t}\Big|_{t=0} \frac{\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} E^tG^t - (F^t)^2}{2\sqrt{EG - F^2}}\,\mathrm{d}u\mathrm{d}v \\ &= -2\int_D \eta H\sqrt{EG - F^2}\,\mathrm{d}u\mathrm{d}v. \qquad \square\end{aligned}$$

Since $EG - F^2 > 0$ if we require that

$$\int_D \eta H\sqrt{EG - F^2}\,\mathrm{d}u\mathrm{d}v = 0$$

for all η it must be that $H = 0$ which yields the following partial differential equation

$$LG - 2MF + NE = 0 \tag{1.2.6}$$

known as the *parametric minimal surface equation.*

Definition 1.2.16. Solutions to Eq. (1.2.6) are known as *parametric minimal surfaces.*

It is straight-forward to check that everything we have done generalizes the previous case of graphs, i.e., if S is a graph then Eq. (1.2.6) reduces to the minimal surface equation for graphs Eq. (1.1.1). It is also the case that minimal surfaces only minimize area locally and not globally, thus all (regular enough) solutions to Plateau's Problem will satisfy Eq. (1.2.6) but not all minimal surfaces will be solutions to Plateau's Problem.

1.3 Minimal surfaces as harmonic functions

We mentioned in §1.1 that the equation of minimal graphs (Eq. (1.1.1)) resembles Laplace's equation when f_x and f_y are very small. This turns out to be just the tip of the iceberg when relating minimal surfaces to harmonic functions and the key to building the connection between the two concepts is an appropriate choice of parameters. For a brief review of harmonic functions, see Appendix A.2.

Definition 1.3.1. A parametrization (D, φ) of a surface S satisfying that φ_u and φ_v are orthogonal $\langle \varphi_u, \varphi_v \rangle = 0$ and have the same moduli $\|\varphi_u\| = \|\varphi_v\|$ is said to define *isothermal coordinates* for S.

In terms of the first fundamental form coefficients, φ determines isothermal parameters if and only if $E = G$ and $F = 0$ which implies that the local expression of the first fundamental form is diagonal

$$[\mathrm{I}_p]_{\mathcal{B}} = \lambda^2 \,\mathrm{Id}_2$$

where $\lambda : D \to \mathbb{R}$ is a smooth function. The defining property of isothermal coordinates is that they preserve angles from the parameter space to the image space. Indeed, for a regular point $p \in S$ let $X_1, X_2 \in \mathbb{R}^2$, $Y_j = d\varphi_p(X_j) \in T_pS$ for $j = 1, 2$ and θ be the angle between X_1 and X_2 and θ' the angle between the image vectors, then

$$\cos\theta' = \frac{\langle Y_1, Y_2 \rangle}{\|Y_1\|\|Y_2\|} = \frac{\lambda^2 \langle X_1, X_2 \rangle}{\lambda\|X_1\|\,\lambda\|X_2\|} = \cos\theta.$$

Definition 1.3.2. A map preserving angles between tangent spaces is known as *conformal.*

Our main interest lies in the fact that, in isothermal coordinates, the mean curvature of a surface can be expressed in terms of the Laplacian of the parametrization.

Proposition 1.3.3. *Let S be a regular surface and (D, φ) a parametrization that defines isothermal coordinates. Then*

$$\Delta\varphi = 2\lambda^2 \vec{H};$$

in particular, S is a minimal surface if and only if the coordinates are harmonic.

Proof. We have that $E = G$ which we can differentiate with respect to u to get

$$\langle \varphi_u, \varphi_{uu} \rangle = \langle \varphi_v, \varphi_{uv} \rangle.$$

Similarly, differentiating $F = 0$ with respect to v we find that

$$\langle \varphi_{uv}, \varphi_v \rangle = -\langle \varphi_u, \varphi_{vv} \rangle$$

which combined with the first identity yields

$$\langle \varphi_u, \varphi_{uu} + \varphi_{vv} \rangle = 0.$$

An analogous procedure gives us that $\langle \varphi_v, \varphi_{uu} + \varphi_{vv} \rangle = 0$ implying that $\varphi_{uu} + \varphi_{vv} \in (T_pS)^\perp = \operatorname{span}(\nu(p))$. Finally, since $E = G = \lambda^2,\ F = 0$

$$H = \frac{LG + NE}{2EG} = \frac{L + N}{2\lambda^2}$$

and we have that

$$\Delta\varphi = \langle \nu, \varphi_{uu} + \varphi_{vv} \rangle \nu = (L + N)\nu = 2\lambda^2 H\nu = 2\lambda^2 \vec{H}.$$ □

We find that Laplace's equation $\Delta\varphi = 0$ defines minimal surfaces, as long as we are working with isothermal coordinates. There are however two issues to be dealt with:

- Is it always the case that there exist isothermal coordinates for a surface?
- Is there a more fundamental relationship between the minimal surface equation and Laplace's equation than just that given in isothermal coordinates?

Each of these questions take us down two different paths within the theory of minimal surfaces both of which are highly relevant. The answer to the first question is affirmative under some general assumptions and it is a key fact in the proof to Plateau's Problem presented in Chapter 2. As shown in §2.2, the existence of isothermal coordinates also highlights one of the many ways in which complex analysis is involved in the theory of minimal surfaces.

As for the second question, we have seen that in a very specific set of coordinates minimal surfaces are described by harmonic functions, but in arbitrary coordinates this is not the case. This leads us to think that we can extend our definition of harmonic functions to somehow encompass non-isothermal parametrizations.

Definition 1.3.4. Let $S = \varphi(D)$ be an immersed surface and $f : S \to \mathbb{R}^k$ a map such that $\hat{f} = f \circ \varphi \in \mathcal{C}^2(D)$. The *Laplace-Beltrami operator* Δ_g acting on f is defined by

$$\Delta_g f := \frac{1}{\sqrt{EG - F^2}} \left[\frac{\partial}{\partial u} \left(\frac{G\hat{f}_u - F\hat{f}_v}{\sqrt{EG - F^2}} \right) + \frac{\partial}{\partial v} \left(\frac{E\hat{f}_v - F\hat{f}_u}{\sqrt{EG - F^2}} \right) \right]. \tag{1.3.1}$$

We say f is *harmonic on* S if $\Delta_g f = 0$.

Applying the Laplace-Beltrami operator on the surface coordinates we find a simple formula for the mean curvature of a surface generalizing Prop. 1.3.3 to general coordinates of S.

Proposition 1.3.5. *Let $S = \varphi(D)$ be an immersed surface, then*

$$\Delta_g x = 2\vec{H}. \tag{1.3.2}$$

where $x : S \to \mathbb{R}^3$ are the coordinate functions $x(p) = p$.

Sketch of Proof. We will have $\hat{x} = \varphi$ such that

$$\Delta_g x = \frac{1}{\sqrt{EG - F^2}} \left[\frac{\partial}{\partial u} \left(\frac{G\varphi_u - F\varphi_v}{\sqrt{EG - F^2}} \right) + \frac{\partial}{\partial v} \left(\frac{E\varphi_v - F\varphi_u}{\sqrt{EG - F^2}} \right) \right]$$

and a tedious calculation shows that $\langle \Delta_g x, \varphi_u \rangle = \langle \Delta_g x, \varphi_v \rangle = 0$ and $\langle \Delta_g x, \hat{\nu} \rangle = 2H$.

As an immediate corollary we have the anticipated characterization of minimal surfaces.

Corollary 1.3.6. *$S = \varphi(D)$ is a minimal surface if and only if its coordinates are harmonic on S.*

At first sight the expression for Δ_g seems complicated, justified only by the fact that it gives the desired result. As a matter of fact, the expression in Eq. (1.3.1) is just a particular case of the more general Laplace-Beltrami operator on manifolds which generalizes the usual Laplacian taking curvature into account. The g in the sub-index references the fact that the operator is constructed from the metric g of a Riemannian manifold $(\mathcal{M}, g)$ which is nothing more than the first fundamental form in a more general setting.

1.4 The connection with complex analysis

Now that we know isothermal coordinates on a minimal surface correspond to harmonic functions, our next goal is to describe such surfaces using holomorphic functions. This approach relies on the classical result that the real and imaginary parts of a holomorphic function are harmonic. For a brief review of complex functions see Appendix A.1.

Definition 1.4.1. Let $S = \varphi(D)$ be a $\mathcal{C}^1$ immersed surface and define the complex function $\Phi : D \to \mathbb{C}^3$, $\Phi = (\phi_1, \phi_2, \phi_3)$ via

$$\phi_j(z) := 2\frac{\partial \varphi^j}{\partial z} = \frac{\partial \varphi^j}{\partial u} - i\frac{\partial \varphi^j}{\partial v} \tag{1.4.1}$$

for $j = 1, 2, 3$. Φ is known as the *Weierstrass-Enneper (WE)* parametrization of S.

The first important property of the WE parametrization is that it allows us to determine minimal surfaces when the ϕ_j's are holomorphic.

Proposition 1.4.2. *Let $\phi_1, \phi_2, \phi_3 : D \to \mathbb{C}$ be holomorphic functions and define $\Phi = (\phi_1, \phi_2, \phi_3)$ and*

$$\varphi^j(u, v) := \operatorname{Re}\left\{\int \phi_j(z)\ \mathrm{d}z\right\} \tag{1.4.2}$$

where $z = u + iv$ and the integral is a complex antiderivative. Then $S = \varphi(D)$ is an immersed minimal surface with φ defining isothermal parameters if and only if the ϕ_j's satisfy

$$\|\Phi\|^2 := \sum_{j=1}^{3} |\phi_j|^2 > 0 \quad \textit{(immersion)} \tag{1.4.3}$$

and

$$\langle \Phi, \Phi \rangle := \sum_{j=1}^{3} (\phi_j)^2 = 0 \quad \textit{(conformality).} \tag{1.4.4}$$

Proof. On one hand we have that

$$\|\Phi\|^2 = \sum_{j=1}^{3} |\phi_j|^2 = \sum_{j=1}^{3} \left(\frac{\partial \varphi^j}{\partial u}\right)^2 + \left(\frac{\partial \varphi^j}{\partial v}\right)^2 = E + G$$

and

$$\langle \Phi, \Phi \rangle = \sum_{j=1}^{3} (\phi_j)^2 = \sum_{j=1}^{3} \left(\frac{\partial \varphi^j}{\partial u}\right)^2 - \left(\frac{\partial \varphi^j}{\partial v}\right)^2 - 2i\left(\frac{\partial \varphi^j}{\partial u}\right)\left(\frac{\partial \varphi^j}{\partial v}\right)^2 = E - G - 2iF$$

so $\langle \Phi, \Phi \rangle = 0$ if and only if $E = G$ and $F = 0$ and $\|\Phi\|^2 > 0$ if and only if $E > 0$ which means φ is an immersion defining isothermal parameters for S.

On the other hand, we can express the Laplacian in terms of the Wirtinger derivatives

$$\frac{\partial}{\partial \overline{z}}\frac{\partial}{\partial z} = \frac{1}{4}\left(\frac{\partial^2}{\partial u^2} + \frac{\partial^2}{\partial v^2}\right) = \frac{1}{4}\Delta$$

so we have that

$$\Delta \varphi^j = \operatorname{Re}\left\{\Delta \int \phi_j(z)\right\} \mathrm{d}z = 4\operatorname{Re}\left\{\frac{\partial}{\partial \overline{z}}\frac{\partial}{\partial z}\int \phi_j(z)\ \mathrm{d}z\right\} = 4\operatorname{Re}\left\{\frac{\partial \phi_j}{\partial \overline{z}}\right\} = 0$$

using the fact that a function satisfies the CR equations if and only if its derivative with respect to $\overline{z}$ vanishes. By Prop. 1.3.3 S must be minimal. $\square$

As we will later see, it is sometimes of interest to allow minimal surfaces which are not necessarily immersed everywhere, for this we make the following definition.

Definition 1.4.3. We say that $S \subset \mathbb{R}^3$ determined by holomorphic ϕ_1, ϕ_2, ϕ_3 according to Eq. (1.4.2) is a *generalized minimal surface* if $\|\Phi\|^2 \geq 0$, $\langle \Phi, \Phi \rangle = 0$ and S contains more than a single point. A point where $\|\Phi\|^2 = 0$ is known as a *branch point* of S.

At its branch points φ is not an immersion, however this cannot happen at many points. Indeed, $\|\Phi\|^2 = 0$ implies that $|\phi_j| = 0$ for $j = 1, 2, 3$ so branch points must be isolated since otherwise the analicity of the ϕ_j's would imply that $\phi_j = 0$ everywhere in D (Prop. A.1.2) implying φ is constant and does not determine a surface. This means that the surface obtained by removing the branch points of a generalized minimal surface is an immersed minimal surface. Allowing for branch points gives us a general method for producing WE parametrizations and thus generalized minimal surfaces.

Proposition 1.4.4. *Let $f : D \to \mathbb{C}$ be a holomorphic function and $g : D \to \mathbb{C}$ be a meromorphic function such that whenever g has a pole of order m then f has a 0 of order at least $2m$, i.e. fg^2 is holomorphic, then*

$$\phi_1 = f(1 - g^2)/2, \quad \phi_2 = if(1 + g^2)/2, \quad \phi_3 = fg \tag{1.4.5}$$

define a WE parametrization of a generalized minimal surface. Conversely, for any WE parametrization Φ of a generalized minimal surface S there exist f and g such that (1.4.5) *holds provided $\phi_1 - i\phi_2$ is not identically 0 in D.*

Proof. For the direct implication we note that since fg^2 is holomorphic in D so are the ϕ_j's and it is easily checked that

$$\sum_{j=1}^{3} (\phi_j)^2 = 0, \quad \sum_{j=1}^{3} |\phi_j|^2 = \frac{|f|^2 (1 + |g|^2)^2}{2} \geq 0$$

so applying a very slightly modified version of Prop. 1.4.2 for generalized minimal surfaces we are done.

For the converse implication, define

$$f := \phi_1 - i\phi_2, \quad \text{and} \quad g := \frac{\phi_3}{\phi_1 - i\phi_2},$$

then clearly $fg = \phi_3$. We also have

$$0 = \phi_1^2 + \phi_2^2 + \phi_3^2 = (\phi_1 - i\phi_2)(\phi_1 + i\phi_2) + \phi_3^2 \iff \phi_1 + i\phi_2 = \frac{-\phi_3^2}{\phi_1 - i\phi_2} = fg^2$$

which tells us that fg^2 is holomorphic and implies together with the definition of f that

$$\phi_1 = f(1 - g^2)/2, \quad \phi_2 = if(1 + g^2)/2. \qquad \square$$

Remark 1.4.5. The condition that $\phi_1 - i\phi_2 \neq 0$ identically is actually necessary. Indeed, if it were the case that $\phi_1 - i\phi_2 = 0$ in D then from the conformality condition we find that $\phi_3 = 0$ everywhere. This would imply that either $f = 0$ or $g = 0$. If $f = 0$ then so would ϕ_1 and ϕ_2 implying $\Phi = 0$ and the WE parametrization determines a single point which does not correspond to a generalized minimal surface. If $g = 0$ then $\phi_1 + i\phi_2 = 0$ which combined with the initial assumption yields again $\phi_1 = \phi_2 = 0$.

Definition 1.4.6. The pair (f, g) in Prop. 1.4.5 are known as the *Weierstrass-Enneper data* of the generalized minimal surface S and

$$\varphi(u, v) = \mathrm{Re}\left\{ \int \frac{f}{2}(1 - g^2, i + ig^2, 2g)\ \mathrm{d}z \right\} \tag{1.4.6}$$

is the *Weierstrass-Enneper representation.*

Remark 1.4.7. A minimal surface's WE data is not necessarily unique, for example choosing $g = 0$ and $f = \alpha$ with $\alpha \in \mathbb{C}$ will always produce the plane $\{z = 0\} \subset \mathbb{R}^3$ for any non zero α.

Although the decomposition of the ϕ_j's in f and g may seem arbitrary there is a clear geometrical interpretation of where they come from.

Proposition 1.4.8. *Let (f, g) be the WE data of an immersed minimal surface S, then $p \circ \hat{\nu} = g$ where p is the stereographic projection. In particular, an immersed surface in isothermal coordinates is minimal if and only if its Gauss map is conformal except possibly at isolated points.*

Proof. We begin by computing the partial derivatives of the parametrization of S associated to (f, g) given by the WE representation

$$\varphi_u = \operatorname{Re}\{\Phi\}, \quad \varphi_v = -\operatorname{Im}\{\Phi\}$$

where we have exchanged the real part operator with the derivative and used that $z = u + iv$. Now, using the definition of $\hat{\nu}$ (Def. 1.2.12) and some simple calculations we find the following expressions for the components of the parametric Gauss map

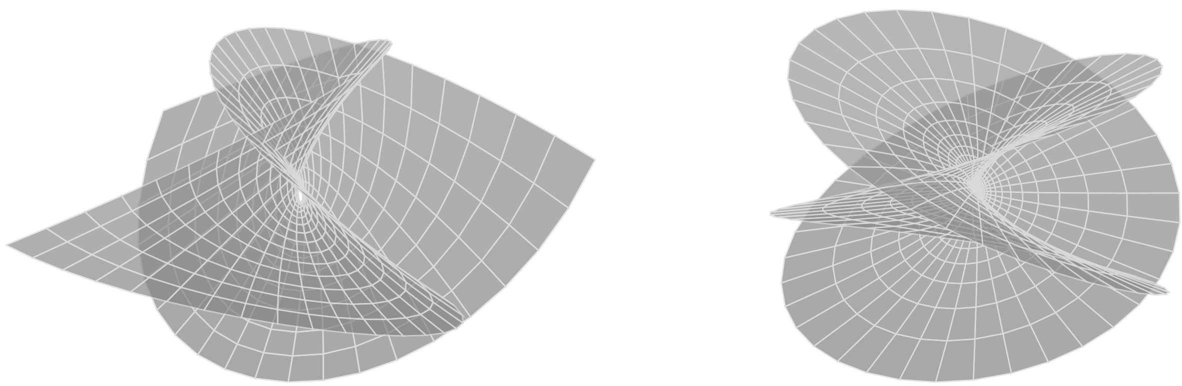
$$\hat{\nu}_1 = -2\operatorname{Im}(\overline{\phi}_2\phi_3)/\|\Phi\|^2\,, \quad \hat{\nu}_2 = -2\operatorname{Im}(\phi_1\overline{\phi}_3)/\|\Phi\|^2\,, \quad \hat{\nu}_3 = -2\operatorname{Im}(\overline{\phi}_1\phi_2)/\|\Phi\|^2$$

where we have used the fact that φ defines isothermal parameters meaning that $E = G = \|\Phi\|^2/2$. Projecting (via (A.1.2)) onto the plane we get

$$\frac{\hat{\nu}_1}{1-\hat{\nu}_3} + \frac{\hat{\nu}_2}{1-\hat{\nu}_3}i = \frac{-2\operatorname{Im}(\overline{\phi}_2\phi_3) - 2\operatorname{Im}(\phi_1\overline{\phi}_3)i}{\|\Phi\|^2 + 2\operatorname{Im}(\overline{\phi}_1\phi_2)} = \frac{|f|^2(1+|g|^2)\operatorname{Re}(g) + |f|^2(1+|g|^2)\operatorname{Im}(g)i}{|f|^2(1+|g|^4+2|g|^2)/2 + |f|^2(1-|g|^4)/2} = g.$$

The direct implication of second part of the statement is a consequence of the fact that meromorphic functions are conformal everywhere except at their poles and zeroes of the derivative, combining this with $\hat{\nu} = p^{-1} \circ g$ it must be that whenever g is conformal so is $\hat{\nu}$ since p is, otherwise the composition need not be conformal. For the converse, if $\hat{\nu}$ is conformal then so is g and if φ defines isothermal parameters then $\Phi = 2\varphi_z$ will have holomorphic components and in particular $f = \phi_1 - i\phi_2$ will also be holomorphic. □

The main consequence of the WE representation of a minimal surface is that it allows the construction of generalized minimal surfaces in isothermal coordinates just from a pair of complex functions satisfying very general conditions. For example, as long as we choose g holomorphic then any holomorphic $f \neq 0$ will give the WE data of an immersed minimal surface. This was a big breakthrough in the study of minimal surfaces in the XIXth century, since up to that time very few examples were known. Exotic examples such as the Enneper surface and the Henneberg surface (Fig. 1.1) were found thanks to the WE representation exhibiting properties that were previously not found in minimal surfaces.



(a) Enneper: $f(z) = 1,\ g(z) = z$

(b) Henneberg: $f(z) = 1 - z^{-4},\ g(z) = z$

Figure 1.1: The Enneper and Henneberg surfaces are two minimal surfaces that can be obtained directly from the WE data. Both are self-intersecting and algebraic, meaning they are the set of roots of a polynomial, with the latter also being non-orientable.

1.5 The catenoid

After going over the basic concepts involved in the study of minimal surfaces it is useful to look at examples. In the following pages we aim to see how the previous concepts apply to a fundamental example of a minimal surface: *the catenoid.*

Definition 1.5.1. Let $I \subseteq \mathbb{R}$ be an open interval and let $\gamma : I \to \mathbb{R}^n$ be a $\mathcal{C}^k$ map, $k \geq 1$. We say that γ is a *regular curve* if $\|\gamma'(t)\| > 0$ for all $t \in I$. A *reparametrization* of γ is a curve $\tilde{\gamma} = \gamma \circ h$, where $h : J \to I$ is a $\mathcal{C}^k$ diffeomorphism onto its image. If $\|\gamma'(s)\| = 1$ for all s, then γ is said to be *parametrized by arc length*, and s is called an *arc-length parameter.*

It is a standard result of the theory of curves in euclidean space that any regular curve has a reparametrization $h(t) = s$ such that s is an arc-length parameter. Regular curves in the plane allow us to construct *surfaces of revolution* by revolving them around a given axis. WLOG the axis of revolution can be assumed to be the z axis up to rigid motions of $\mathbb{R}^3$ and in this case a parametrization is given by

$$\begin{aligned} \varphi : (a,b) \times \mathbb{R} &\to \mathbb{R}^3 \\ (u,v) &\mapsto (h(u)\cos v, h(u)\sin v, g(u)) \end{aligned}$$

where we assume that the curve $(h(u), g(u))$ is $\mathcal{C}^2$, regular and $h > 0$ to ensure that φ is an immersion. If we suppose that u is an arc-length parameter, then a direct computation shows that the expressions for the Gauss and mean curvature are the following.

$$K = \frac{g'h'(g''h' - g'h'')}{h^2}, \quad H = \frac{h^2(g''h' - g'h'') + g'h}{2h^2}. \tag{1.5.1}$$

We note that the expressions for K and H are independent of the angular coordinate v, as expected from the symmetry of the surfaces. As a consequence, the equation for minimal surfaces $H = 0$ reduces to a second-order ordinary differential equation, which can be solved using elementary methods to find that there are only two minimal surfaces of revolution.

Proposition 1.5.2. *The plane $\{z = 0\} \subset \mathbb{R}^3$ and the immersed surface parametrized by*

$$\begin{aligned} \varphi : \mathbb{R} \times \mathbb{R} &\to \mathbb{R}^3 \\ (u,v) &\mapsto (a\cosh\big(u/a\big)\cos\big(v/a\big), a\cosh\big(u/a\big)\sin\big(v/a\big), u), \end{aligned} \tag{1.5.2}$$

known as the catenoid[1]*, are the only minimal surfaces of revolution up to reparametrizations and rigid motions of $\mathbb{R}^3$, where $a > 0$.*

Sketch of Proof. The proof consists in solving $H = 0$ in Eq. (1.5.1) for g and h, treating separately the cases where $g' = 0$ at some point and where it is never zero. Uniqueness follows from the standard uniqueness results for solutions to ODEs. For the full proof, see Prop. 12.2.2 of [Pre10].

The plane is obtained by revolving a straight line and the catenoid is obtained by revolving the graph of $f(x) = a\cosh\big(x/a\big)$ which is known as the *catenary* curve. The later curve is the shape that a piece of loose rope adopts when tied between two separate vertical poles and is left hanging under its own weight. A notable fact about the catenary is that its shape can be obtained from applying variational methods to the gravitational potential energy of the rope with the aim to minimize such energy. This is analogous to what the catenoid does in 2D in the case that we are trying to minimize the surface tension of a soap film or bubble joining two circles.

[1]As presented in Eq. (1.5.2) the curve which is revolved to form the catenoid is not parameterized by arc-length.

Now, using the formulas from §1.2 we can compute the parametric first fundamental form and Gauss map of the catenoid

$$[\mathrm{I}_p]_{\mathcal{B}} = \cosh^2(u/a)\,\mathrm{Id}_2, \quad \nu = \frac{1}{\cosh(u/a)}(-\cos(v/a), -\sin(v/a), \sinh(u/a))$$

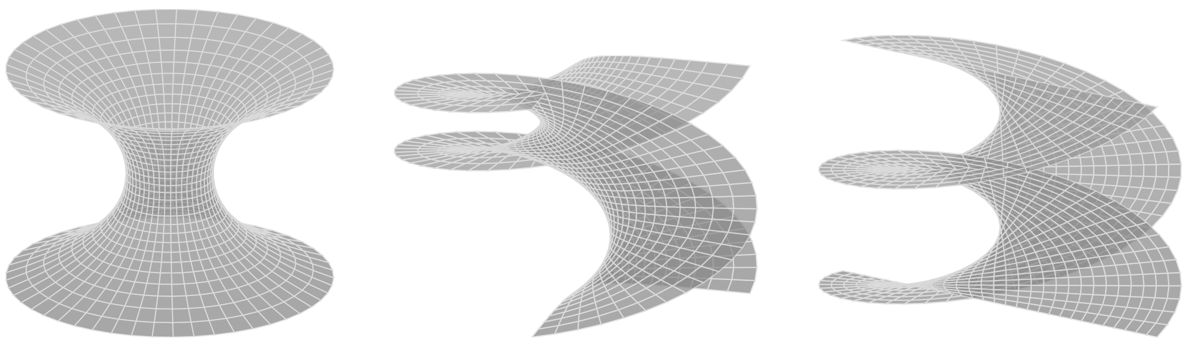
and we see that the parametrization of Eq. (1.5.2) defines isothermal coordinates. This means we can use Prop. 1.4.8 to find the WE data of the catenoid

$$g = p(\nu) = \frac{-\cos(v/a)}{\cosh(u/a) - \sinh(u/a)} + \frac{-\sin(v/a)}{\cosh(u/a) - \sinh(u/a)} i = \frac{-e^{iv/a}}{\cosh(u/a) - \sinh(u/a)}$$

and using the definitions of the hyperbolic trig functions we find that $\sinh u - \cosh u = -e^{-u}$ so for $z = u + iv$ we have

$$g(z) = -e^{z/a}$$

which is a very beautiful result. Now, f can be determined using the fact that $\phi_3 = fg$ and computing $\phi_3 = 2\varphi_z^3 = 1$ we find that $f(z) = -e^{-z/a}$ so the pair $(-e^{-z/a}, -e^{z/a})$ is the WE data of the catenoid. Historically, the catenoid was the first non-trivial minimal surface discovered in 1744 by Euler. In 1776 Jean Baptiste Meusnier discovered the helicoid which is the second non-trivial minimal surface. By adding a phase factor to f such that $f(z;\theta) = e^{i\theta}e^{-z/a}$, with $\theta \in \mathbb{R}$, the pair $(f(z;\theta), g(z))$ corresponds to the WE data of a 1 parameter family of minimal surfaces with the catenoid sitting at $\theta \in \pi\mathbb{Z}$. For $\theta \in \pi/2 + \pi\mathbb{Z}$ we obtain the helicoid and varying θ we obtain a smooth transformation between the helicoid and the catenoid. Such a family of minimal surfaces is known as an *associate family* or a *Bonnet family* and whenever two minimal surfaces in the same family differ by $\Delta\theta = \pi/2$ they are called *conjugate*.



(a) Catenoid ($\theta = 0$) (b) Helicatenoid ($\theta = \pi/4$) (c) Helicoid ($\theta = \pi/2$)

Figure 1.2: Bonnet family corresponding to the WE data $f(z;\theta) = e^{i\theta}e^{-z}$ and $g(z) = -e^z$.

To finish this brief study of the catenoid we solve the following Plateau Problem:

Given two identical circles one on top of the other separated by some distance in $\mathbb{R}^3$*, is there a surface that is bounded by them with minimal area?*

To proceed we suppose that we are dealing with two circles of unit radius along the z axis and separated a distance of $2h$ for some $h > 0$ fixed

$$C_\pm := \{(x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 = 1,\ z = \pm h\}.$$

Without going into formalities, it is clear that any surface that is area minimizing between these two circles must have rotational symmetry around the z axis since in the contrary we could take an infinitesimally small section of the surface having the least area among all such sections and rotate it around to get a surface of revolution with less area. Because of this and the result in Prop. 1.5.2 the only possible solutions to this Plateau Problem are a horizontal strip of a catenoid bounded by $C_+ \cup C_-$ or the two disks

$$D_\pm := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 \leq 1,\ z = \pm h\}.$$

Whether the answer will be the disks or the catenoid depends on which has the least area, which for the disks is clearly always 2π. The case of the catenoid is a bit more delicate since, as we will see now, there are values of h for which no catenoid contains the circles $C_\pm$ and values for which there are two distinct such catenoids. Due to the symmetry, this problem can be studied by simply looking at the profile, this means we can take $v = 0$ and we just need to require that

$$a \cosh\big(h/a\big) = 1 \tag{1.5.3}$$

for a catenoid that contains $C_\pm$ to exist. Thus the existence of catenoids whose boundary are unit circles at height $\pm h$ is equivalent to the existence of solutions to the previous equation. Attempting to find a as function of h analytically is not very fruitful so we will settle with a numerical approach. In Fig. 1.3 we can see that for $h > h_{\text{crit}} \approx 0.66274$ there are no values of a satisfying the equation which means that if the circles are separated a distance larger than $2h_{\text{crit}}$ the only minimal surface bounding them will be $D_+ \cup D_-$ with an area of 2π. Now, if $h = h_{\text{crit}}$ there exists a single catenoid with a parameter value of $a_{\text{crit}} \approx 0.55243$. The area of a horizontal strip of a catenoid going from $z = -h$ to $z = h$ is given by

$$A(h) = \int_0^{2\pi} \int_{-h}^{h} \cosh^2(u/a)\ \mathrm{d}u\mathrm{d}v = 4\pi a \int_0^{h/a} \cosh^2(t)\ \mathrm{d}t = 2\pi a \Big[\cosh\big(h/a\big) \sinh\big(h/a\big) + h/a\Big],$$

so we will have $A(h_{\text{crit}}) \approx 13.64467 > 2\pi$ meaning that for the critical height the Plateau Problem still has the disks as its solution with an area considerably smaller than the case of the catenoid. Finally, we look at the case of $h < h_{\text{crit}}$. From Fig. 1.3 we see that as h falls below its critical value there are two branches for a which we call a_l and a_r (for left and right) such that $0 < a_l < a_{\text{crit}} < a_r < 1$. We notice that using the condition in Eq. (1.5.3) we can rewrite the area as

$$A(h) = 2\pi \left[\sqrt{\frac{1}{a^2} - 1} + h\right]$$

which shows that the area increases as a decreases to 0 corresponding to the branch a_l. On the right branch, as h goes to 0^+ we have $a_r \to 1^-$ and this makes the area decrease to 0; therefore, Bolzano's theorem implies that there exists $0 < h_{2\pi} < h_{\text{crit}}$ defined implicitly by $A(h_{2\pi}) = 2\pi$. Numerically, we find that

$$h_{2\pi} \approx 0.45661, \quad a_{r,2\pi} \approx 0.87866$$

so for $h < h_{2\pi}$ the solution to the Plateau Problem will be the catenoid with parameter $a_r(h)$. In Fig. 1.4 we can see why the catenary corresponding to a_l has a larger area than that of a_r for the same h. There are several interesting takeaways from this example:

- We have verified that there are minimal surfaces that are not solutions to Plateau's Problem, i.e., locally but no globally minimizing area.
- We have found that for $h = h_{2\pi}$ the area of the minimizing catenoid is the same as that of the disks, so both are solutions to Plateau's Problem and we don't have uniqueness. Furthermore, each solution has a different topology and in particular one is connected while the other is not.
- In relation to the dependence on h, we have found that there is a discontinuity at $h = h_{2\pi}$ where we

jump from a catenoid to the disks. This means that we cannot expect in general that the solution to a continuous parametric family of Plateau Problems to depend continuously on the parameter.

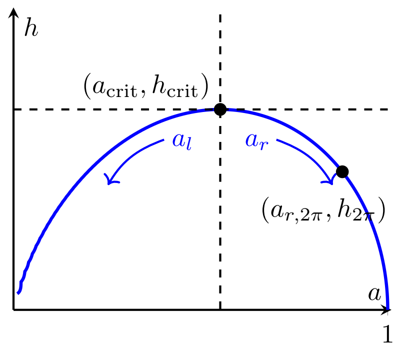


Figure 1.3: Implicit plot of the curve $a\cosh(h/a) = 1$ for $h > 0$, relating the allowed values of a for each h, and the points $(a_{\text{crit}}, h_{\text{crit}})$ and $(a_{r,2\pi}, h_{2\pi})$.

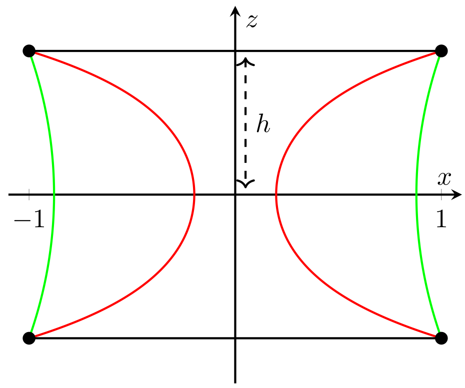


Figure 1.4: Profile of the two possible catenoids for $h = h_{2\pi}$, in green for a_t and in red for a_b. The black lines between the black dots are the circles.

For the general case of two disks of radius R we see that Eq. (1.5.3) turns to $a\cosh(h/a) = R$ which can be rewritten as $a'\cosh(h'/a') = 1$ where $a' = a/R$ and $h' = h/R$ so the analysis will be exactly the same up to a scaling factor. Finally, we can give the answer to the Plateau Problem:

There always exists an area minimizing surface that bounds two identical circles lying on on top of the other for any distance separating them. Furthermore, there is a concrete distance between the two circles for which there exists two surfaces of distinct topology that minimize the area.

2 Douglas's solution to Plateau's Problem

In this chapter we present an adapted version of the proof by Douglas which simplifies his original argument, spanning a total of 59 pages [Dou31]. We begin by discussing the precise formulation of the problem to be solved. Before moving on to the proof, we take a deeper look into the existence of isothermal coordinates for surfaces since it is an essential result for what follows. We have split the main proof into a creative part and a technical part. The creative part is concerned with reformulating the problem so that it becomes more approachable, while the technical part deals with the subtleties of solving the reformulated problem. In the end, we combine both parts to give the full proof of Plateau's Problem, and in the last section we discuss how this version of the proof both differs from and resembles Douglas's original argument. This version of the proof is based on the contents of Chapter 4 in [Xin18] which in its turn is based on a 1937 paper by Courant [Cou37] following up on Douglas's proof.

2.1 On the precise formulation

We have already seen in §1.5 that specifying the boundary of a surface is not restrictive enough to determine the topology of the area-minimizing surface that spans it. However, the example of the two circles is not of the type that we will be focusing on, since the boundary is disconnected. Regardless, there also exist examples of a single connected bounding curve admitting different area-minimizing surfaces in the topological sense. Although Plateau's Problem, per se, is not concerned with the uniqueness of solutions, it is very helpful to restrict our attention to classes of surfaces with fixed topology, the simplest of which is the disk $\mathbb{D}$. After all, it is natural that before considering surfaces with more complex topology, such as those having holes, one would first ask whether the simplest solution exists. Because of this, we will also restrict the bounding curves to the following class.

Definition 2.1.1. $\Gamma \subset \mathbb{R}^n$ is said to be a *Jordan curve* if it is homeomorphic to the circle $\partial\mathbb{D} = \mathbb{S}^1$.

We note that we are allowing curves that may be knotted implying they only bound surfaces having self-intersections. The solution of Plateau's Problem that we will present does not care for this and the question asking whether a given solution is embedded (regular) is a different one. Still, when considering how a parametrization maps onto the minimal surface solution we will need to be careful with how it parametrizes Γ.

Definition 2.1.2. A continuous map $\gamma : \partial\mathbb{D} \to \Gamma$ is *monotonic* if $\gamma^{-1}(\{p\})$ is connected for every $p \in \Gamma$.

In essence, a monotonic map is one that always moves in the same direction, this means that γ is allowed to stop at certain points so it need not be injective but it can never go backwards. We choose this property on the parametrization of the bounding curves because, unlike injectivity, it is preserved under limits.

Regarding the smoothness of the surface parametrizations, we will require that they be $\mathcal{C}^1$ in $\mathbb{D}$ for the area to be well defined. With all this, given a Jordan curve Γ the class of surfaces having this boundary that we will admit are those parametrized by functions in

$$X_\Gamma := \{\varphi : \overline{\mathbb{D}} \to \mathbb{R}^n \mid \varphi \text{ is continuous and } \mathcal{C}^1 \text{ in } \mathbb{D} \text{ and } \varphi|_{\partial\mathbb{D}} : \partial\mathbb{D} \to \Gamma \text{ is monotonic and surjective}\}.$$

As before, we define the area of the surface determined by a parametrization φ according to Eq. (1.1.2). More precisely, we make the following definition.

Definition 2.1.3. The *area functional* $A : X_\Gamma \to [0, +\infty]$ is defined by

$$A(\varphi) = \int_{\mathbb{D}} \sqrt{\|\varphi_u\|^2 \|\varphi_v\|^2 - \langle \varphi_u, \varphi_v \rangle^2}\, \mathrm{d}u\mathrm{d}v = \int_{\mathbb{D}} \sqrt{EG - F^2}\, \mathrm{d}u\mathrm{d}v.$$

The final important assumption that we need to make is that there exist a surface spanning Γ with a finite area. This is not always the case and Douglas gives an example at the end of his paper by defining a continuous but very irregular curve. Another such example was constructed by Besicovitch in 1955 [Bes55]. A sufficient condition to ensure that Γ admits a surface of finite area is that it is rectifiable, i.e. of finite length.

Definition 2.1.4 (Plateau Problem)**.** The *classical Plateau Problem* for $\mathcal{C}^1$ surfaces in $\mathbb{R}^n$ is the question asking whether there exists a parametrization $\varphi \in X_\Gamma$ minimizing the area functional A, that is

$$A(\varphi) = \mathcal{A}_\Gamma := \inf_{\psi \in X_\Gamma} A(\psi),$$

provided $\mathcal{A}_\Gamma < +\infty$.

It is important to note that X_Γ makes no mention of the immersion property of the parametrizations, so apriori the surface solutions to Plateau's Problem need not be regular or even immersed. Additionally, we are only demanding the parametrizations to be $\mathcal{C}^1$ so their mean curvature may not be defined even if the surface is regular. Luckily, as we will later see these issues (almost) sort themselves out as a bonus from the proof. Nevertheless, it is important to keep in mind that we begin without assuming any additional properties of the parametrizations as we did throughout Chapter 1.

Ideally, to solve Plateau's Problem we would proceed as follows. By the definition of infimum there must exist a sequence of parametrizations $\{\varphi_k\}_k$ such that $A(\varphi_k) \to A_\Gamma$ as $k \to \infty$. We would then try to extract a convergent subsequence $\{\varphi_k\}_k$ such that the limit is the desired solution. It turns out that the area functional carries with it two important flaws that obstruct this procedure:

- We don't have necessarily that $\varphi_k \to \varphi$ uniformly for some $\varphi \in X_\Gamma$ or even for a subsequence $\varphi_{k_p} \to \varphi$. To see this consider the non-converging family of diffeomorphisms of the disk $\phi_k : \overline{\mathbb{D}} \to \overline{\mathbb{D}}$ for $k \in \mathbb{N}$ given by $\phi_k(z) = e^{ik}z$ then $\tilde{\varphi}_k = \varphi_k \circ \phi_k$ has $A(\tilde{\varphi}_k) = A(\varphi_k)$, since the area functional is invariant under reparametrizations, but $\{\tilde{\varphi}_k\}_k$ is not necessarily convergent so it could be that $\{\varphi_k\}_k$ is already of this form and also non-convergent. This problem stems from the fact that the group of diffeomorphisms of the disk $\mathbb{D}$ is not compact with respect to the uniform norm and A is reparametrization invariant.
- The second problem is that $\{\varphi_k\}_k$ could be convergent but with a limit that is not in X_Γ as shown in Fig. 2.1. This is equivalent to saying that X_Γ is not closed with respect to the uniform norm.

In conclusion, the area functional is not directly minimizable. On the one hand, it is invariant under reparametrizations, while we are interested in working with an explicit, convergent sequence of parametrizations. On the other hand, controlling the area of the images tells us very little about the behavior of the parametrizations themselves. Both these problems are rooted in the fact that X_Γ is not compact. Therefore, the first step of the proof will be to look for another functional that is better behaved for our purposes. The full proof can be split into two pieces:

(a) The «creative» piece: find a functional that is more robust and show that minimizing this new functional is equivalent to minimizing the area functional in some sense.

(b) The «technical» piece: prove the existence of minimizers for the new functional using the techniques of functional analysis.

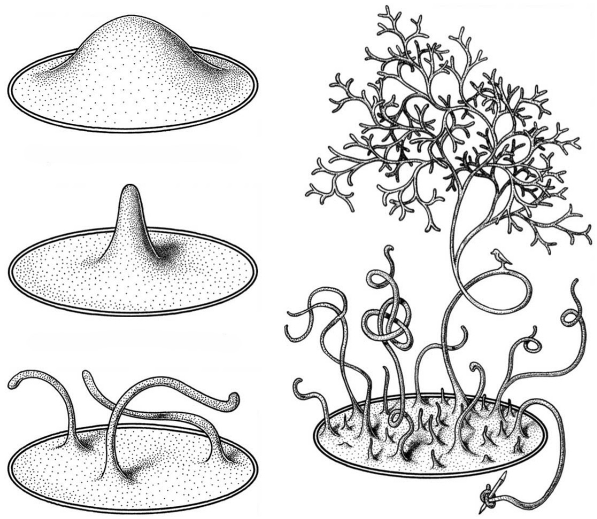


Figure 2.1: A wildly non-converging sequence of surfaces in X_Γ whose area converges to $\mathcal{A}_\Gamma$ for $\Gamma = \mathbb{S}^1 \subset \mathbb{R}^3$. Illustration by James F. Bredt (Source: [Mor08]).

2.2 More on isothermal coordinates

Before proceeding, a further discussion on isothermal coordinates is warranted as their existence will be essential in converting our study of the area functional into the study a more appropriate functional. Our goal now is to take a brief look into what exactly is required for a surface to have isothermal parameters which ideally won't be much so as to not have to add any major restrictions to our statement of Plateau's Problem.

In the following it will be more convenient to work in complex coordinates. We can formally take $(z, \overline{z})$ as coordinates in $\mathbb{R}^2$ using their relation to the standard coordinates

$$\begin{cases} z = u + iv \\ \overline{z} = u - iv \end{cases} \iff \begin{cases} u = \frac{z+\overline{z}}{2} \\ v = \frac{z-\overline{z}}{2i} \end{cases}$$

and define the Wirtinger derivatives ∂_z and $\partial_{\overline{z}}$ such that the chain rule holds. We can extend this procedure to use parametrizations of surfaces in complex coordinates as follows. Suppose $\varphi : \mathbb{D} \to \mathbb{R}^n$ is a $\mathcal{C}^1$ immersion and consider $\psi(z, \overline{z})$ the change to complex coordinates from $\varphi(u, v)$, then

$$\psi_z = \frac{1}{2}(\varphi_u - i\varphi_v) \quad \psi_{\overline{z}} = \frac{1}{2}(\varphi_u + i\varphi_v),$$

and taking inner products we find

$$\langle \psi_z, \psi_z \rangle = \frac{E - G - 2iF}{4}, \quad \langle \psi_z, \psi_{\overline{z}} \rangle = \frac{E + G}{4}, \quad \langle \psi_{\overline{z}}, \psi_{\overline{z}} \rangle = \frac{E - G + 2iF}{4}, \quad .$$

We formally define the coefficients of the first fundamental form in complex coordinates and its matrix as follows

$$\alpha := \frac{E - G + 2iF}{4} \in \mathbb{C}, \quad \beta := \frac{E + G}{4} > 0; \quad [\mathrm{I}_p]_{(z,\overline{z})} = \begin{pmatrix} \overline{\alpha} & \beta \\ \beta & \alpha \end{pmatrix}.$$

We then have that $E = G$ and $F = 0$ if and only if $\alpha = 0$ so complex isothermal parameters are those having zeroes in the diagonal of the first fundamental form matrix. We are now interested in finding new coordinates $w = f(z, \overline{z})$ where f is a $\mathcal{C}^1$ diffeomorphism such that $\alpha = 0$ in the new coordinates $(w, \overline{w})$.

We emphasize the dependence of f on $\overline{z}$ since in general it will not be holomorphic because if it were then it would be conformal and that would imply $(z,\overline{z})$ are already isothermal. Suppose $\tilde{\psi}(w,\overline{w})$ is an isothermal parametrization and let $\psi(z,\overline{z}) = \tilde{\psi}(f(z,\overline{z}), \overline{f(z,\overline{z})})$, then we compute

$$\psi_z = \tilde{\psi}_w w_z + \tilde{\psi}_{\overline{w}} \overline{w}_z, \quad \psi_{\overline{z}} = \tilde{\psi}_w w_{\overline{z}} + \tilde{\psi}_{\overline{w}} \overline{w}_{\overline{z}}$$

which we can invert to find

$$\tilde{\psi}_w = \frac{1}{w_z \overline{w}_{\overline{z}} - \overline{w}_z w_{\overline{z}}} (\psi_z \overline{w}_{\overline{z}} - \psi_{\overline{z}} \overline{w}_z), \quad \tilde{\psi}_{\overline{w}} = \frac{1}{w_z \overline{w}_{\overline{z}} - \overline{w}_z w_{\overline{z}}} (\psi_{\overline{z}} w_z - \psi_z w_{\overline{z}})$$

and compute the conjugate of the α coefficient in the new coordinates

$$\overline{\alpha^w} = \frac{\alpha w_z^2 - 2\beta w_z w_{\overline{z}} + \overline{\alpha} w_{\overline{z}}^2}{(w_z \overline{w}_{\overline{z}} - \overline{w}_z w_{\overline{z}})^2}.$$

If $(w,\overline{w})$ are isothermal the previous quantity must vanish so w must satisfy the following equation

$$\alpha w_z^2 - 2\beta w_z w_{\overline{z}} + \overline{\alpha} w_{\overline{z}}^2 = 0,$$

and dividing[2] by w_z yields

$$\alpha - 2\beta\mu + \overline{\alpha}\mu^2 = 0$$

where $\mu = w_{\overline{z}}/w_z$. Solving the previous equation with the quadratic formula we find

$$\mu = \frac{\beta \pm \sqrt{\beta^2 - |\alpha|^2}}{\overline{\alpha}} = \frac{E + G \pm 2\sqrt{EG - F^2}}{E - G - 2iF},$$

and we have the following result.

Proposition 2.2.1. *The reparametrization $w = f(z,\overline{z})$ defines isothermal coordinates if and only if w satisfies the following partial differential equation*

$$\mu_\pm \frac{\partial w}{\partial z} = \frac{\partial w}{\partial \overline{z}}$$

where $\mu_\pm$ is defined by

$$\mu_\pm := \frac{E + G \pm 2\sqrt{EG - F^2}}{E - G - 2iF}.$$

Furthermore, the Jacobian determinant of the transformation is

$$\frac{\partial(w,\overline{w})}{\partial(z,\overline{z})} = |f_z|^2 (1 - |\mu_\pm|^2),$$

in particular $|\mu_-| < 1$ implies f is orientation preserving and $|\mu_+| > 1$ implies f reverses orientation.

Proof. We have already seen the first part and for the second we first compute the differential matrix

$$Df = \begin{pmatrix} f_z & f_{\overline{z}} \\ \overline{f}_z & \overline{f}_{\overline{z}} \end{pmatrix}.$$

[2] Since f is a diffeomorphism is must be injective everywhere implying its differential is always of rank 2, this has the consequence that $w_z \neq 0$ everywhere.

From the definition of the Wirtinger derivatives it can be seen that $\overline{f}_z = \overline{f_{\overline{z}}}$ and $\overline{f}_{\overline{z}} = \overline{f_z}$ such that

$$\det(Df) = |f_z|^2 - |f_{\overline{z}}|^2 = |f_z|^2\,(1 - |\mu_\pm|^2).$$

Finally, for μ_- we have

$$|\mu_-|^2 = \frac{\left(E + G - 2\sqrt{EG - F^2}\right)^2}{(E-G)^2 + 4F^2};$$

and expanding the numerator

$$(E + G - 2\sqrt{EG - F^2})^2 = E^2 + 2EG + G^2 - 4(E+G)\sqrt{EG - F^2} + 4(EG - F^2)$$

and the denominator

$$(E-G)^2 + 4F^2 = E^2 - 2EG + G^2 + 4F^2.$$

So $|\mu_-|^2 < 1$ if and only if

$$8EG - 4(E+G)\sqrt{EG - F^2} < 8F^2 \iff 2\sqrt{EG - F^2} < E + G$$

but the last inequality is true precisely because we are assuming our original coordinates are not isothermal. Proving $|\mu_+| > 1$ is analogous. □

We are particularly interested in the case of f preserving the orientation.

Definition 2.2.2. The *Beltrami coefficient* is defined as $\mu := \mu_-$ and

$$\mu \frac{\partial w}{\partial z} = \frac{\partial w}{\partial \overline{z}} \tag{2.2.1}$$

is known as the *Beltrami equation.*

The Beltrami equation was first introduced by Gauss in the XIXth century to show the existence of local isothermal coordinates for analytic first fundamental forms. Since then, it has been extensively studied as it plays a very important role in complex analysis, differential geometry of surfaces, Teichmüller theory or PDEs in the plane. For our purposes, we are interested in the existence solutions that are global $\mathcal{C}^1$ diffeomorphisms of the disk $\mathbb{D}$. The existence of such solutions is guaranteed mainly by the *Measurable Riemann Mapping Theorem* which ensures a global homeomorphic solution to Eq. (2.2.1) provided $|\mu| \leq k < 1$. The regularity of solutions to the Beltrami equation can be upgraded to $\mathcal{C}^1$ by applying elliptic regularity results if μ is locally α-Hölder continuous[3] for $0 < \alpha < 1$. For a complete treatment of both results see [AIM09], namely Theorem 5.3.2 and Theorem 15.6.2. We note that in our current formulation the hypothesis of these theorems are not met, in particular we are not sure of the behavior of $|\mu|$ close to $\partial\mathbb{D}$; a sufficient condition for $|\mu| \leq k < 1$ in $\mathbb{D}$ is that

$$\inf_{(u,v)\in\mathbb{D}} \frac{\sqrt{EG - F^2}}{E + G} > 0. \tag{2.2.2}$$

We will also need to require that E, F and G are locally α-Hölder continuous such that μ is. For a derivation of the condition in Eq. (2.2.2) and a more in depth discussion on the technical hypothesis on the existence of an isothermal reparametrization see Appendix C.

[3] A function $f : U \to \mathbb{R}$ is *α-Hölder continuous* if there exists $C > 0$ such that $|f(x) - f(y)| \leq C|x - y|^\alpha$ for all $x, y \in U$.

2.3 The «creative» piece

In the introduction we looked at how minimal surfaces are the higher dimensional equivalent of geodesics, which minimize the length between two points of a space. It is a well-known fact from physics that free falling objects through curved spacetime minimize along geodesics minimize their kinetic energy. Motivated by this analogy and the problems we have discussed related to the area functional we define the following alternative functional.

Definition 2.3.1. For $\varphi \in X_\Gamma$ we define its *Dirichlet energy* by

$$E(\varphi) := \frac{1}{2}\int_{\mathbb{D}} \|\nabla\varphi\|^2 \, \mathrm{d}u\mathrm{d}v := \frac{1}{2}\int_{\mathbb{D}} \|\varphi_u\|^2 + \|\varphi_v\|^2 \, \mathrm{d}u\mathrm{d}v = \frac{1}{2}\int_{\mathbb{D}} E + G \, \mathrm{d}u\mathrm{d}v,$$

and analogously to the area case we define

$$\mathcal{E}_\Gamma = \inf_{\varphi\in X_\Gamma} E(\varphi).$$

Remark 2.3.2. Note that we are defining the norm of the gradient of a vector valued function as $\|\nabla\varphi\|^2 = \|\varphi_u\|^2 + \|\varphi_v\|^2$, corresponding to the norm of $J\varphi$ thought of as a vector in $\mathbb{R}^{2n}$.

The Dirichlet energy avoids the second problem we had with the area functional since surfaces like the ones in Fig. 2.1 necessarily have big variations making $\|\nabla\varphi\|$ large and in consequence $E(\varphi)$. As for the parametrization invariance problem, it may seem at first that we have also solved it since $E(\varphi)$ is not reparametrization invariant, but the following result shows that we have some work left to do.

Proposition 2.3.3. *The Dirichlet energy of a function is conformal invariant.*

Proof. A conformal reparametrization is of the form $w = f(z)$ with $f : \mathbb{D} \to \mathbb{D}$ a holomorphic diffeomorphism. Letting $w = u + iv$ and $z = x + iy$ and $\tilde{\varphi} = \varphi \circ f$ we have

$$\begin{aligned} 4\left\|\frac{\partial\tilde{\varphi}}{\partial z}\right\|^2 &= \|\tilde{\varphi}_x\|^2 + \|\tilde{\varphi}_y\|^2 = \|\varphi_u u_x + \varphi_v v_x\|^2 + \|\varphi_u u_y + \varphi_v v_y\|^2 \\ &= \|\varphi_u\|^2 (u_x^2 + u_y^2) + \|\varphi_v\|^2 (v_x^2 + v_y^2) + \langle\varphi_u, \varphi_v\rangle(u_x v_x + u_y v_y). \end{aligned}$$

Now, since f is holomorphic its components satisfy the Cauchy-Riemann equations which gives

$$u_x^2 + u_y^2 = v_x^2 + v_y^2 = u_x v_y - u_y v_x = \frac{\partial(u,v)}{\partial(x,y)} \quad \text{and} \quad u_x v_x + u_y v_y = -v_y u_y + u_y v_y = 0.$$

All in all,

$$\int_{\mathbb{D}} \|\tilde{\varphi}_x\|^2 + \|\tilde{\varphi}_y\|^2 \, \mathrm{d}x\mathrm{d}y = \int_{\mathbb{D}} \left(\|\varphi_u\|^2 + \|\varphi_v\|^2\right) \frac{\partial(u,v)}{\partial(x,y)} \, \mathrm{d}x\mathrm{d}y = \int_{\mathbb{D}} \|\varphi_u\|^2 + \|\varphi_v\|^2 \, \mathrm{d}u\mathrm{d}v. \qquad \square$$

So the Dirichlet energy still has invariance with respect to a smaller class of reparametrizations of the disk which is also non-compact. Despite this, the reduction on the size of allowable reparametrizations is enough that we will be able to circumvent it in a way that will actually help us with the proof.

Now, we note that for two vectors $X, Y \in \mathbb{R}^n$ we have

$$\|X\|^2\|Y\|^2 - \langle X, Y\rangle^2 \le \|X\|^2\|Y\|^2 \le \frac{1}{4}\left(\|X\|^2 + \|Y\|^2\right)^2 \tag{2.3.1}$$

with equality if and only if $\langle X, Y\rangle^2 = 0$ and $\|X\| = \|Y\|$, meaning that for any $\varphi \in X_\Gamma$

$$A(\varphi) \leq E(\varphi).$$

and equality holds if and only if $\|\varphi_u\| = \|\varphi_v\|$ and $\langle \varphi_u, \varphi_v\rangle = 0$ everywhere in $\mathbb{D}$. Maps satisfying these conditions are known as *weakly conformal* given that they are conformal maps but dropping the positivity on $\|\varphi_u\| = \|\varphi_v\| \geq 0$ which allows for branch points on our surfaces. Despite E being larger than A, they achieve the same infimum value on X_Γ.

Lemma 2.3.4. *Suppose $\{\varphi_k\}_k$ is a sequence in X_Γ with $A(\varphi_k) \to \mathcal{A}_\Gamma$ with each φ_k satisfying condition (2.2.2) with locally $\mathcal{C}^{0,\alpha_k}$ first fundamental form coefficients for some $0 < \alpha_k < 1$, then $\mathcal{E}_\Gamma = \mathcal{A}_\Gamma$.*

Proof. The $\geq$ inequality is a consequence of $A(\varphi) \leq E(\varphi)$. For the converse inequality, we would like for the φ_k to define isothermal parameters but since we cannot be sure that they are immersions we have to perform a work around. For this let $\epsilon > 0$ and define $\varphi_{k,\epsilon} : \overline{\mathbb{D}} \to \mathbb{R}^{n+2}$ by

$$\varphi_{k,\epsilon}(u, v) = (\varphi_k(u, v), \epsilon u, \epsilon v)$$

which has first fundamental form coefficients

$$E_{k,\epsilon} = E_k + \epsilon^2, \quad F_{k,\epsilon} = F_k, \quad G_{k,\epsilon} = G_k + \epsilon^2$$

such that $\varphi_{k,\epsilon}$ is an immersion for any ϵ. We have that

$$A(\varphi_{k,\epsilon}) = \int_{\mathbb{D}} \sqrt{(E_k + \epsilon^2)(G_k + \epsilon^2) - F_{k,\epsilon}^2}\, \mathrm{d}u\mathrm{d}v = \int_{\mathbb{D}} \sqrt{E_k G_k - F_k^2 + \epsilon^2(E_k + G_k) + \epsilon^4}\, \mathrm{d}u\mathrm{d}v$$

and condition (2.2.2) guarantees that $\int_{\mathbb{D}} E_k + G_k \leq \int_{\mathbb{D}} \sqrt{E_k G_k - F_k^2} < \infty$. As a consequence, for each k we can choose ϵ small enough such that

$$A(\varphi_k) \leq A(\varphi_{k,\epsilon}) < A(\varphi_k) + \frac{1}{k}.$$

Now, $\varphi_{k,\epsilon}$ are immersions with locally $\mathcal{C}^{0,\alpha}$ first fundamental form coefficients and (it is easily verified that) they satisfy condition (2.2.2). Then, we know that for each k there exists a reparametrization $f_{k,\epsilon} : \mathbb{D} \to \mathbb{D}$ such that $(\tilde{u}, \tilde{v}) = f_{k,\epsilon}^{-1}(u, v)$ are isothermal parameters for the surface defined by $\varphi_{k,\epsilon}$ so

$$\tilde{\varphi}_{k,\epsilon}(\tilde{u}, \tilde{v}) = (\varphi_{k,\epsilon} \circ f_{k,\epsilon})(\tilde{u}, \tilde{v}) = \big((\varphi_k \circ f_{k,\epsilon})(\tilde{u}, \tilde{v}), \epsilon u(\tilde{u}, \tilde{v}), \epsilon v(\tilde{u}, \tilde{v})\big)$$

will be conformal. We then have that

$$E(\varphi_k \circ f_{k,\epsilon}) \leq E(\tilde{\varphi}_{k,\epsilon}) = A(\tilde{\varphi}_{k,\epsilon}) = A(\varphi_{k,\epsilon}) < A(\varphi_k) + \frac{1}{k},$$

where the first inequality is consequence of the triangle inequality, the second equality is by the conformality of the reparametrized $\varphi_{k,\epsilon}$ and the third by invariance of A under reparametrizations. Finally,

$$\mathcal{E}_\Gamma \leq \liminf_{k\to\infty} E(\varphi_k \circ f_{k,\epsilon}) \leq \lim_{k\to\infty} A(\varphi_k) + \frac{1}{k} = \mathcal{A}_\Gamma.$$ □

Remark 2.3.5. The technical assumptions in the previous Lemma are there to ensure that there exists a diffeomorphic reparametrization that yields isothermal parameters; however, it is not clear when we can ensure they are satisfied. A sufficient condition for condition (2.2.2) to be met is that Γ be $\mathcal{C}^1$ such that the parametrizations are regular up to the boundary $\varphi \in \mathcal{C}^1(\overline{\mathbb{D}})$. As for the locally $\mathcal{C}^{0,\alpha_k}$ first fundamental form coefficients, this can be ensured by increasing the required regularity in $\mathbb{D}$ of the parametrizations. However, this is not necessary as in general a more regular sequence whose area converges to the infimum can be constructed from one that is only $\mathcal{C}^1$.

Finally, we can show that a minimizer of E is also a minimizer of A.

Proposition 2.3.6. *$\varphi \in X_\Gamma$ satisfies $E(\varphi) = \mathcal{E}_\Gamma$ if and only if $A(\varphi) = \mathcal{A}_\Gamma$ and φ is weakly conformal.*

Proof. If $E(\varphi) = \mathcal{E}_\Gamma$ then by the previous lemma

$$A(\varphi) \le E(\varphi) = \mathcal{E}_\Gamma = \mathcal{A}_\Gamma \le A(\varphi)$$

so we have a chain of equalities and $A(\varphi) = \mathcal{A}_\Gamma$ and we also have $A(\varphi) = E(\varphi)$ which implies that φ is weakly conformal. Conversely, if $A(\varphi) = \mathcal{A}_\Gamma$ then again by the previous lemma $A(\varphi) = \mathcal{E}_\Gamma$ and if φ is weakly conformal then $A(\varphi) = E(\varphi)$ so $E(\varphi) = \mathcal{E}_\Gamma$. □

With this, our problem can be reformulated as the question on the existence of a minimizer of the Dirichlet energy among the class X_Γ. Luckily, we are already very familiar with the minimizers of E when we fix the boundary condition to a concrete parametrization of Γ.

Proposition 2.3.7 (Poisson's Formula)**.** *Fix $g : \partial\mathbb{D} \to \mathbb{R}$ continuous and define the Poisson kernel $P : \mathbb{D} \to [0, +\infty)$ by*

$$P(r, \theta) := \frac{1 - r^2}{1 - 2r\cos(\theta) + r^2}.$$

Then the function $u : \overline{\mathbb{D}} \to \mathbb{R}$ defined by

$$u(re^{i\theta}) = \begin{cases} \frac{1}{2\pi}\int_0^{2\pi} P(r, \theta - \tau) g(e^{i\tau})\ \mathrm{d}\tau, & \text{if } r < 1 \\ g(e^{i\theta}), & \text{if } r = 1 \end{cases}$$

is the unique classical solution to the Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } \mathbb{D}, \\ u = g & \text{in } \partial\mathbb{D}; \end{cases}$$

with $u \in \mathcal{C}^\infty(\mathbb{D}) \cup \mathcal{C}^0(\overline{\mathbb{D}})$.

For a proof of Poisson's formula and a more detailed treatment of harmonic functions, the Poisson kernel and the Dirichlet problem see Appendix A.2. The key result for our case is that solutions to a given Dirichlet problem are minimizers of the Dirichlet energy with the same boundary condition, this is known as *Dirichlet's principle.*

Proposition 2.3.8 (Dirichlet's principle)**.** *Let $\gamma : \partial\mathbb{D} \to \Gamma$ continuous and*

$$X_\gamma = \{\varphi : \overline{\mathbb{D}} \to \mathbb{R}^n \mid \varphi \text{ is continuous and } \mathcal{C}^1 \text{ in } \mathbb{D} \text{ and } \varphi|_{\partial\mathbb{D}} = \gamma\}.$$

If

$$\mathcal{E}_\gamma := \inf_{\varphi \in X_\gamma} E(\varphi) < \infty$$

then there exists a unique $\varphi_\gamma \in X_\gamma$ such that $E(\varphi_\gamma) = \mathcal{E}_\gamma$. Moreover, φ_γ is $\mathcal{C}^\infty$ and harmonic in $\mathbb{D}$.

Proof. For $j = 1, \ldots, n$ let $\varphi_\gamma^j : \overline{\mathbb{D}} \to \mathbb{R}$ be the unique solution to the boundary problem

$$\begin{cases} \Delta u = 0, \\ u|_{\partial\mathbb{D}} = \gamma^j; \end{cases}$$

given by Poisson's formula. For $\psi \in X_\gamma$ arbitrary, integrating by parts we find

$$0 = \int_{\mathbb{D}} \Delta\varphi_\gamma^j(\psi^j - \varphi_\gamma^j)\, \mathrm{d}u\mathrm{d}v = -\int_{\mathbb{D}} \langle \nabla\varphi_\gamma^j, \nabla(\psi^j - \varphi_\gamma^j)\rangle\, \mathrm{d}u\mathrm{d}v$$

since $\Delta\varphi_\gamma^j = 0$ and $(\psi - \varphi_\gamma)|_{\partial\mathbb{D}} = 0$. We then have that

$$\begin{aligned} 2E(\varphi_\gamma^j) &= \int_{\mathbb{D}} \langle \nabla\varphi_\gamma^j, \nabla\varphi_\gamma^j\rangle\, \mathrm{d}u\mathrm{d}v = \int_{\mathbb{D}} \langle \nabla\varphi_\gamma^j, \nabla\psi^j\rangle\, \mathrm{d}u\mathrm{d}v \\ &\leq \int_{\mathbb{D}} \left\|\nabla\varphi_\gamma^j\right\| \left\|\nabla\psi^j\right\| \mathrm{d}u\mathrm{d}v \leq \frac{1}{2}\int_{\mathbb{D}} \left\|\nabla\varphi_\gamma^j\right\|^2 \mathrm{d}u\mathrm{d}v + \frac{1}{2}\int_{\mathbb{D}} \left\|\nabla\psi^j\right\|^2 \mathrm{d}u\mathrm{d}v = E(\varphi_\gamma^j) + E(\psi^j) \end{aligned}$$

where we have used the inner product Cauchy-Schwarz inequality and the inequality $2ab \leq a^2 + b^2$. Comparing the first and last terms we find that $E(\varphi_\gamma^j) \leq E(\psi^j)$ and in consequence $E(\varphi_\gamma) \leq E(\psi)$ where $\varphi_\gamma : \overline{\mathbb{D}} \to \mathbb{R}^n$ has φ_γ^j as its components. We find that φ_γ is unique and, by construction, harmonic with $\varphi_\gamma \in X_\gamma$ which completes the proof. □

Remark 2.3.9. A reader familiar with Dirichlet's principle but not with the theory of minimal surfaces might have already guessed that minimizing the Dirichlet energy is an equivalent approach, once the link between minimal surfaces and harmonic functions is established in Prop. 1.3.3. This is of course the original motivation for introducing the Dirichlet energy, which had been recognized years before Douglas's proof.

2.4 The «technical» piece

To complete the proof, all we have left to do is minimize the Dirichlet energy on X_Γ for which we already know that for each parametrization of the boundary $\gamma : \partial\mathbb{D} \to \Gamma$ there exists a unique minimizer $\varphi_\gamma \in X_\Gamma$ given by the Poisson integral. Despite this, it still might be that for each γ the value of $\mathcal{E}_\gamma = E(\varphi_\gamma)$ is different so we are left to show that there is one such that $\mathcal{E}_\gamma = \mathcal{E}_\Gamma$.

We start by dealing with the small problem we left hanging in the previous section, namely the fact that the Dirichlet integral is conformal invariant. To solve this we choose three different arbitrary fixed points $z_1, z_2, z_3 \in \partial\mathbb{D}$ and three other different points $p_1, p_2, p_3 \in \Gamma$ and define the following subclass of parametrizations

$$X_\Gamma^* := \{\varphi \in X_\Gamma \mid \varphi(z_j) = p_j \text{ for } j = 1, 2, 3\},$$

for which we have the following result.

Proposition 2.4.1. *Let*

$$\mathcal{E}_\Gamma^* := \inf_{\varphi \in X_\Gamma^*} E(\varphi),$$

then $\mathcal{E}_\Gamma^* = \mathcal{E}_\Gamma$.

Proof. From $X_\Gamma^* \subseteq X_\Gamma$ we immediately find that $\mathcal{E}_\Gamma^* \geq \mathcal{E}_\Gamma$ since we are searching among a larger class of minimizers. For the reverse inequality, let $\{\varphi_k\}_k$ be a sequence of functions in X_Γ such that $E(\varphi_k) \to \mathcal{E}_\Gamma$ as $k \to \infty$ and for each k let $w_{k,j} \in \partial\mathbb{D}$ be such that $\varphi_k(w_{k,j}) = p_j$ for $j = 1, 2, 3$. Now, for each

k there exists a Möbius transformation f_k mapping the disk to the disk and such that $f_k(z_j) = w_{k,j}$. Define $\varphi_k^* := \varphi_k \circ f_k$ which will be in X_Γ^* and since f_k is conformal we will have by Prop. 2.3.3 that $E(\varphi_k^*) = E(\varphi_k)$ so $\lim_k E(\varphi_k^*) = \mathcal{E}_\Gamma$ which implies $\mathcal{E}_\Gamma^* \leq \mathcal{E}_\Gamma$. □

Remark 2.4.2. The fact that we fixed three points is no coincidence as we know that the conformal reparametrizations of the extended complex plane will be Möbius transformations which are determined by the image of three points. For the disk the most general form of a conformal reparametrization will be

$$f(z) = e^{i\theta} \frac{z-a}{1-\bar{a}z}$$

for $\theta \in \mathbb{R}$ and $a \in \mathbb{D}$ corresponding to a total of three real parameters. In the previous proof, these parameters get determined by the arguments of the z_j's.

This tells us that we can focus our attention on finding minimizers in the class X_Γ^*. The fixing of three points has a very important added bonus which is that the smaller class X_Γ^* has the property that the set of parametrizations of Γ it induces by taking the restriction $\varphi|_{\partial\mathbb{D}}$ for $\varphi \in X_\Gamma^*$ is (almost) compact. Before stating this result, we remind ourselves of the well-known Arzelà-Ascoli theorem which is a key part of the proof and tells us that sequences of functions with certain continuity properties must have a convergent subsequence.

Definition 2.4.3. Let (X, d) be a metric space and $\mathcal{F} = \{f_\alpha : X \to \mathbb{R}\}$ a family of real valued functions. $\mathcal{F}$ is said to be:

(a) *Uniformly bounded* if there exists a constant $M > 0$ such that $\left|f_\alpha(x)\right| \leq M$ for every $x \in X$ and $f_\alpha \in \mathcal{F}$.

(b) *Uniformly equicontinuous* if for every $\epsilon > 0$ there exists $\delta > 0$ such that for any $x, y \in X$ with $d(x, y) < \delta$ then $\left|f_\alpha(x) - f_\alpha(y)\right| < \epsilon$ for any $f_\alpha \in \mathcal{F}$.

Theorem 2.4.4 (Arzelà-Ascoli)**.** *Let (X, d) be a compact metric space and $f_m : X \to \mathbb{R}$ be a uniformly bounded and uniformly equicontinuous sequence of functions. Then there exists a uniformly convergent subsequence f_{m_k}.*

Proof. See Theorem 11.28 of [Rud74]. □

We are now ready to address the most technical part of the entire proof. For this, it will be useful to introduce the following notation: for any $r > 0$ and $z \in \mathbb{C}$, we define the set

$$C_r(z) := \{w \in \overline{\mathbb{D}} \mid |w - z| = r\},$$

that is, $C_r(z)$ is the portion of the circle of radius r centered at z that lies inside the closed unit disk.

The next series of results are aimed at establishing the compactness properties of X_Γ^*. We begin by deriving a well-known bound on the arc-length of the image of $C_r(z)$ under a map $\varphi \in X_\Gamma$ known as the *Courant-Lebesgue Lemma*. This estimate will then be used to prove that subsets of X_Γ^* with uniformly bounded Dirichlet energy are equicontinuous. Finally, combining these results with a few technical observations will allow us to invoke the Arzelà–Ascoli theorem and conclude the existence of a convergent subsequence within X_Γ^*.

Lemma 2.4.5 (Courant-Lebesgue Lemma)**.** *Let $\varphi \in X_\Gamma$ and $C > 0$ a constant with $E(\varphi) \leq M$. Then for any $z \in \mathbb{C}$ and $0 < \delta < 1$ there exists ρ with $\delta \leq \rho \leq \sqrt{\delta}$ such that*

$$\ell(\varphi(C_\rho(z))) \leq \sqrt{2\pi\kappa(\delta)}$$

where $\ell(\varphi(C_\rho(z)))$ is the length of the curve determined by the image of $C_\rho(z)$ through φ and

$$\kappa(\delta) = \frac{2M}{\log\bigl(1/\delta\bigr)}.$$

Proof. Fix $z \in \mathbb{C}$, $r > 0$ and let s be the arc length parameter of $C_r(z)$, i.e., such that $\mathrm{d}s = r\mathrm{d}\theta$. One can easily verify from the form of the gradient in polar coordinates[4] that

$$\|\nabla\varphi\|^2 = \|\varphi_r\|^2 + \|\varphi_s\|^2 \geq \|\varphi_s\|^2$$

which then implies that

$$\int_\delta^{\sqrt{\delta}} \int_{C_r(z)} \|\varphi_s\|^2 \ \mathrm{d}s\mathrm{d}r \leq \int_{\mathbb{D}} \|\nabla\varphi\|^2 \,\mathrm{d}u\mathrm{d}v = E(\varphi) \leq M.$$

Now, since φ is $\mathcal{C}^1$ in $\overline{\mathbb{D}}$ its derivative is continuous and we can apply the mean value theorem for integrals[5] to get

$$\begin{aligned}\int_\delta^{\sqrt{\delta}} \int_{C_r(z)} \|\varphi_s\|^2 \ \mathrm{d}s\mathrm{d}r &= \int_\delta^{\sqrt{\delta}} \left(r \int_{C_r(z)} \|\varphi_s\|^2 \ \mathrm{d}s \right) \frac{1}{r} \ \mathrm{d}r \\ &= \rho \int_{C_\rho(z)} \|\varphi_s\|^2 \ \mathrm{d}s \int_\delta^{\sqrt{\delta}} \frac{1}{r} \ \mathrm{d}r = \rho \int_{C_\rho(z)} \|\varphi_s\|^2 \ \mathrm{d}s \,\frac{\log\bigl(1/\delta\bigr)}{2}\end{aligned}$$

for some $\rho \in \left[\delta, \sqrt{\delta}\right]$ and combining this with the previous inequality we find

$$\int_{C_\rho(z)} \|\varphi_s\|^2 \ \mathrm{d}s \leq \frac{\kappa(\delta)}{\rho}.$$

To finish the proof of the lemma, we use the previous inequality and the integral version of the Cauchy-Schwarz inequality

$$\ell(\varphi(C_\rho(z)))^2 = \left(\int_{C_\rho(z)} \|\varphi_s\| \ \mathrm{d}s \right)^2 \leq \left(\int_{C_\rho(z)} \|\varphi_s\|^2 \ \mathrm{d}s \right) \left(\int_{C_\rho(z)} \ \mathrm{d}s \right) \leq \frac{\kappa(\delta)}{\rho} 2\pi\rho = 2\pi\kappa(\delta). \qquad \square$$

Proposition 2.4.6. *Suppose $M > 0$ is a constant such that $\mathcal{E}_\Gamma < M$. Then the family*

$$\mathcal{F} = \{\varphi|_{\partial\mathbb{D}} \mid \varphi \in X_\Gamma^* \text{ and } E(\varphi) \leq M\}$$

is equicontinuous on $\partial\mathbb{D}$.

Proof. Let $\epsilon > 0$. We claim that there exists $\eta > 0$ such that for any distinct points $q, q' \in \Gamma$ with $\|q - q'\| < \eta$ there is at least one connected component of $\Gamma \setminus \{q, q'\}$ having diameter less than ϵ. Indeed, since Γ is homeomorphic to the circle, for each $p \in \Gamma$ there exists an open ball B_p such that $B_p \cap \Gamma$ is connected; we can also assume each B_p to have radius less than $\epsilon/2$. Clearly the B_p's cover Γ so by compactness we find that there are a finite amount of such balls covering the curve. Choosing $\eta = \epsilon/2$ it must be that q and q' live in the same open ball which will also contain one whole part of Γ joining q and q' since we assumed such balls to maintain connectedness. Since this path component is entirely

[4]In polar coordinates $\nabla f = \partial_r f e_r + r^{-1}\partial_\theta f e_\theta = \partial_r f e_r + \partial_s f e_\theta$ where e_r and e_θ are the radial and angular unit vectors, respectively.

[5]Mean Value Theorem for Integrals with a weight function: If f is continuous on $[a, b]$ and g is integrable and non-negative on $[a, b]$, then there exists $c \in [a, b]$ such that $\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx$.

contained in a ball of radius ϵ its diameter will have to be less than or equal to ϵ as claimed.

Using this fact our aim now is to apply the inequality in Lemma 2.4.5 to show the equicontinuity, see Fig. 2.2. Let $z \in \partial\mathbb{D}$ and for ϵ and η the same as before we choose $0 < \delta < 1$ small enough such that $\sqrt{2\pi\kappa(\delta)} < \eta$ and $|z - z_k| > \sqrt{\delta}$ for at least two of the points z_1, z_2 and z_3. We also suppose WLOG that $\epsilon < \min_{i\neq j}\|p_i - p_j\|$. Since $\delta < 1$ the previous lemma tells us that for any $\varphi \in X_\Gamma^*$ there exists $\rho \in \left[\delta, \sqrt{\delta}\right]$ such that

$$\ell(\varphi(C_\rho(z))) \leq \sqrt{2\pi\kappa(\delta)} < \eta.$$

$C_\rho(z)$ will divide $\partial\mathbb{D}$ into a circular arc A containing z and its complementary arc B. Let $\varphi(A)$ and $\varphi(B)$ be the images of A and B in Γ and q, q' the points in Γ separating $\varphi(A)$ and $\varphi(B)$, then

$$\|q - q'\| \leq \ell(\varphi(C_\rho(z))) < \eta$$

so either $\varphi(A)$ or $\varphi(B)$ have diameter less than ϵ. We claim that it is $\varphi(A)$ that has its diameter less than ϵ. Indeed, since $\rho < \sqrt{\delta}$ and from how we chose δ such that $|z - z_k| > \sqrt{\delta}$ for at least two of z_1, z_2 and z_3 we have that $\varphi(B)$ contains two of the p_j's implying

$$\operatorname{diam} \varphi(B) \geq \min_{i\neq j}\|p_i - p_j\| > \epsilon.$$

Finally if $z' \in \partial\mathbb{D}$ is such that $|z' - z| < \delta$ we will have $z' \in A$ (because $\delta < \rho$) implying $\varphi(z') \in \varphi(A)$ and thus

$$\|\varphi(z') - \varphi(z)\| < \operatorname{diam} \varphi(A) < \epsilon,$$

since δ does not depend on the choice of z or φ we have shown that $\mathcal{F}$ is uniformly equicontinuous. $\square$

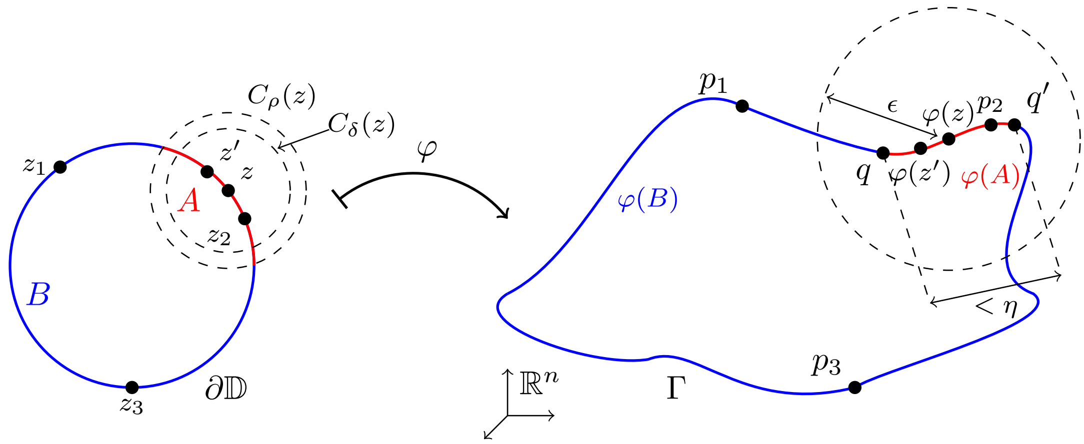


Figure 2.2: Sketch of the situation in the proof of Prop. 2.4.6.

Corollary 2.4.7. *If $\{f_m\}_m$ is a sequence of functions in $\mathcal{F}$ (with $\mathcal{F}$ as in Prop. 2.4.6) then there exists a subsequence f_{m_k} uniformly converging to some $f \in \mathcal{F}$.*

Proof. To apply Arzelà-Ascoli we only have left to show that $\mathcal{F}$ is uniformly bounded but that is immediate if we remember the fact that all its functions map to Γ which is the continuous image of a compact set (the circle). Now, if we have a sequence of functions $\{f_m\}_m$ in $\mathcal{F}$ the Arzelà-Ascoli theorem guarantees the existence of a uniformly convergent subsequence to some function f and we just need to check that it indeed belongs to $\mathcal{F}$.

Since f is the uniform limit of functions in $\mathcal{F}$ it will be monotonic and surjective. We will also have that f is continuous so we can then extend it to the disk using the Poisson kernel to get some harmonic

$\varphi^f : \overline{\mathbb{D}} \to \mathbb{R}^n$. We already know that φ^f will be smooth in $\mathbb{D}$ so in particular it will be $\mathcal{C}^1$ and $\varphi^f \in X_\Gamma$. It is also easy to check that the three points stay fixed $f(z_j) = p_j$ for $j = 1, 2, 3$ since this is a property that is trivially preserved under limits thus $\varphi^f(z_j) = p_j$. Finally, φ^f will have finite Dirichlet energy and we can assume WLOG that $E(\varphi^f) \leq M$ (by possibly choosing a larger M). All in all, we have found that $f \in \mathcal{F}$. □

Remark 2.4.8. In general the Arzelà-Ascoli theorem does not guarantee the convergence inside of $\mathcal{F}$. As shown at the end of the previous proof, verifying that the limiting function is in $\mathcal{F}$ is straightforward in this case.

2.5 Putting the pieces together

At this point we already have all of the ingredients for the proof of the main result and what's left is just a matter of combining them. But before doing so we will have to quickly show one last property of the Dirichlet energy.

Lemma 2.5.1. *Let $f_k : \overline{\mathbb{D}} \to \mathbb{R}$ be a sequence continuous functions in $\overline{\mathbb{D}}$ and harmonic in $\mathbb{D}$ uniformly convergent to f, then*

$$E(f) \leq \liminf_{k\to\infty} E(f_k).$$

Proof. Let $K_m \subseteq \mathbb{D}$ be a sequence of compacts increasing to $\mathbb{D}$, since the f_k are harmonic for each m we will have that $\nabla f_k \to \nabla f$ uniformly on K_m (see Lemma A.2.10). We will then have

$$\int_{\mathbb{D}} \|\nabla f_k\|^2 \, \mathrm{d}u\mathrm{d}v \geq \int_{K_m} \|\nabla f_k\|^2 \, \mathrm{d}u\mathrm{d}v \overset{k\to\infty}{\longrightarrow} \int_{K_m} \|\nabla f\|^2 \, \mathrm{d}u\mathrm{d}v$$

where we have exchanged the integral and the limit using the uniform convergence. We will then have

$$\liminf_{k\to\infty} \int_{\mathbb{D}} \|\nabla f_k\|^2 \, \mathrm{d}u\mathrm{d}v \geq \int_{K_m} \|\nabla f\|^2 \, \mathrm{d}u\mathrm{d}v$$

and letting the compacts tend to the disk $K_m \nearrow \mathbb{D}$ we find

$$\liminf_{k\to\infty} E(f_k) = \liminf_{k\to\infty} \int_{\mathbb{D}} \|\nabla f_k\|^2 \, \mathrm{d}u\mathrm{d}v \geq \int_{\mathbb{D}} \|\nabla f\|^2 \, \mathrm{d}u\mathrm{d}v = E(f).$$ □

The previous property is known as the *lower semi-continuity* of E for harmonic functions. As a matter of fact, the Dirichlet energy is lower semi-continuous on a much broader class of functions but the harmonic case has a much simpler proof and it is all we will need. Having said this, we may finally state and prove the classical Plateau Problem.

Theorem 2.5.2 (Douglas, 1931). *Let $\Gamma \subset \mathbb{R}^n$ be a Jordan curve with $\mathcal{E}_\Gamma < \infty$. Then there exists $\varphi : \overline{\mathbb{D}} \to \mathbb{R}^n$ continuous such that*

(a) $\varphi|_{\partial\mathbb{D}} : \partial\mathbb{D} \to \Gamma$ is monotonic;

(b) φ is harmonic and weakly conformal in the interior of $\mathbb{D}$;

(c) $A(\varphi) = \mathcal{A}_\Gamma$.

Proof. Let $\{\varphi_k\}_k$ be a sequence in X_Γ^* such that $E(\varphi_k) \to \mathcal{E}_\Gamma$ as $k \to \infty$ and we assume WLOG that $\{E(\varphi_k)\}_k$ is a decreasing sequence and $E(\varphi_1) < M < \infty$ so that $\gamma_k := \varphi_k|_{\partial\mathbb{D}} \in \mathcal{F}$. We will then have by Corollary 2.4.7 that there exists a map $\gamma \in \mathcal{F}$ and a subsequence $\{\gamma_{k_m}\}_m$ such that $\gamma_{k_m} \to \gamma$ uniformly. By Dirichlet's principle (Prop. 2.3.8), for each m there exists $\psi_m \in X_\Gamma^*$ such that $E(\psi_m) = \mathcal{E}_{\gamma_{k_m}}$ and

$\psi_m|_{\partial\mathbb{D}} = \gamma_{k_m}$ which is also harmonic. Since each ψ_m is harmonic they must converge uniformly to a harmonic $\psi : \overline{\mathbb{D}} \to \mathbb{R}^n$ satisfying $\psi|_{\partial\mathbb{D}} = \gamma$ (by Prop. A.2.9). Finally, we will have that

$$E(\psi) \leq \liminf_{m\to\infty} E(\psi_m) = \liminf_{m\to\infty} \mathcal{E}_{\gamma_{k_m}} = \lim_{k\to\infty} \mathcal{E}_{\gamma_k} = \mathcal{E}_\Gamma$$

where the first inequality is obtained applying the previous lemma to each of the components of ψ.

Now, since $\psi \in X^*_\Gamma$ its restriction to $\partial\mathbb{D}$ is monotonic which proves point (a). Since $E(\psi) = \mathcal{E}_\Gamma$ by Prop. 2.3.6 we have $A(\varphi) = \mathcal{A}_\Gamma$ proving (c) and ψ is weakly conformal proving (b). ☐

We end by pointing out some consequences of the proof. At the beginning, we only required the surfaces to be once differentiable, but as we have seen, the solutions we obtain are much more regular. On the one hand, any solution φ is automatically $\mathcal{C}^\infty$ in the interior of the disk. On the other, since the solutions are harmonic they are also real analytic. Furthermore, since φ is weakly conformal the surface it defines is a generalized minimal surface whose mean curvature is well defined and vanishes everywhere except at its branch points. Finaly, analicity implies that any compact subset of the disk can contain at most finitely many branch points. In fact, it was shown by Osserman [Oss70] that for solutions in $\mathbb{R}^3$, no branch points appear in the interior at all. This is not true in general for maps into $\mathbb{R}^n$ or for possible branch points along the boundary.

Remark 2.5.3. The version of Douglas's theorem which we have stated is true, however it does not follow directly from our arguments as the hypothesis in Theorem 2.5.2 do not directly imply the conditions of Lemma 2.3.4 are met. For our arguments to hold we can further assume that Γ is $\mathcal{C}^1$ which could then be used to obtain the general case by approximating Γ with smooth functions.

2.6 Douglas's original proof

Although the proof of Plateau's Problem we have presented preserves the core ideas proposed by Douglas in his original argument, there are some differences worth mentioning. The most notable is that Douglas does not minimize the Dirichlet energy in his proof, but rather a different functional whose minimum coincides with that of the area functional. Specifically, Douglas considered the following functional:

$$\mathfrak{A}(g) = \frac{1}{4\pi}\int_0^{2\pi}\int_0^{2\pi} \frac{\sum_{j=1}^n [g_j(e^{i\alpha}) - g_j(e^{i\beta})]^2}{4\sin^2(\frac{\alpha-\beta}{2})}\, \mathrm{d}\alpha \mathrm{d}\beta,$$

where $g : \partial\mathbb{D} \to \Gamma$ is a homeomorphism parameterizing the bounding Jordan curve and g_j are its components.

At first glance, Douglas's functional may seem more complicated than the Dirichlet energy, but this is not because he was unaware of the latter. In fact, Douglas mentions in his paper the criticism made by Weierstrass towards Riemann's use of the Dirichlet energy to solve Plateau's Problem, pointing out that it lacks certain compactness properties which Douglas's functional satisfies. Specifically, the class of functions admissible by the Dirichlet energy is neither compact nor closed. On the other hand, since $\mathfrak{A}$ is only defined on homeomorphisms of the boundary of the disk (rather than on functions over the whole disk), the corresponding class of admissible functions forms a «Fréchet L-set»[6] meaning it becomes compact and closed once we include "improper topological representations" of Γ, i.e., limits of proper ones such as non-injective monotone mappings. This is the role that Corollary 2.4.7 plays in the present proof. Moreover, Douglas's functional is lower semi-continuous which allows the application of a result

[6]The notion of L-sets was first introduced by Maurice Fréchet in the early 20th century in the context of the calculus of variations. Although the term is no longer commonly used today, the concept essentially corresponds to a set of admissible functions that is compact and closed in a suitable function space, ensuring that lower semi-continuous functionals attain their minima over it.

of Fréchet stating that a lower semi-continuous functional on a compact and closed class attains its minimum. This part of the proof is handled in the first half of Douglas's paper and corresponds to the technical piece (§2.4) of this chapter. Another remarkable aspect of Douglas's functional is that it does not involve any derivatives in its integrand, in the words of Courant [Cou37]:

> This ingenious departure from classical lines to a variational problem not implying derivatives makes it easy to establish the existence of a minimizing [parametrization]. Thereby the complications in the method are shifted to the task of excluding harmful singularities from the solution and of identifying the solution with the boundary values of a [parametrization] satisfying [$E - G = F = 0$].

To relate Douglas's functional to the area functional, for any admissible g of $\mathfrak{A}$ we define its harmonic extension to the whole disk via the Poisson kernel. We denote this extension by $\varphi^g : \overline{\mathbb{D}} \to \mathbb{R}^n$ and define its area as

$$A(g) := A(\varphi^g) = \int_{\mathbb{D}} \sqrt{EG - F^2}\ \mathrm{d}u\mathrm{d}v,$$

where E, F and G are the first fundamental form coefficients of φ^g. We then define the complex functions ϕ_j as in Eq. (1.4.1), associated to φ^g. Via an application of Green's theorem, Douglas shows that the functional takes the form

$$\mathfrak{A}(g) = \frac{1}{2}\int_{\mathbb{D}} \sum_{j=1}^{n} \left|\phi_j(w)\right|^2\ \mathrm{d}u\mathrm{d}v = \frac{1}{2}\int_{\mathbb{D}} E + G\ \mathrm{d}u\mathrm{d}v$$

where the second equality follows from the calculation in Prop. 1.4.2 and $w = u + iv$. The conclusion is that, when g is extended harmonically $\mathfrak{A}(g)$ is the Dirichlet energy. Thus $\mathfrak{A}(g) \geq A(g)$ and equality holds if and only if $E = G$ and $F = 0$. In terms of ϕ_j, this condition becomes

$$\sum_{j=1}^{n} \phi_j^2 = 0$$

which holds if and only if φ^g is minimal. Recall that for harmonic parametrizations, minimality is equivalent to isothermality of the parameters. Therefore, to complete the proof, it remains to show that such a reparametrization exists. The process of showing the existence of an isothermal reparametrization differs from ours as Douglas uses an argument of Radó which applies the «conformal mapping theorem of Koebe» to first prove it for a polyhedral surface bounded by Γ from which the conclusion can be extended with a limiting argument. All of this work is carried out in its majority in the second half of Douglas's paper, which in the present adaption would correspond to the creative piece (§2.3) of this chapter.

We thus conclude that the primary difference between the two proofs lies in the choice of functional to minimize, raising the natural question:

Why is it now possible to use the Dirichlet energy?

The reason is subtle, and becomes clear when considering the point at which isothermal parameters are introduced in each proof. In our approach, isothermal coordinates are introduced at the very beginning, to show that E and A attain the same minimum. To do this, we use the powerful Measurable Riemann Mapping Theorem. However, this theorem did not appear until 1938, and its complete proof wasn't available until 1960, which explains why Douglas and Radó could not rely on it. In contrast, Douglas only requires isothermal parameters at the end, when showing that the minimizing g also determines a minimal surface that minimizes area globally. At that stage, the existence of isothermal parameters can be established using the Riemann mapping theorem (at that time referred to as Koebe's due to his proof), which is a special case of the Uniformization Theorem. Proven in 1907 independently by Poincaré and Koebe, the Uniformization theorem[7] states that every simply connected Riemann surface is conformally equivalent to either the unit disk, the complex plane, or the Riemann sphere. We arrive to the conclusion that the tools of complex analysis and the theory of conformal maps are essential in studying the Plateau Problem for surfaces. This already hints that when studying minimal surfaces in higher dimensions, the previous approach may not be as fruitful, since such surfaces generally cannot be parametrized using complex functions.

[7] See [Sai16] for a complete treatment on the Uniformization theorem including the original papers of Poincaré and Koebe.

3 Further directions and closing remarks

The solution to Plateau's Problem given by Theorem 2.5.2 guarantees that for any Jordan curve $\Gamma \subset \mathbb{R}^n$, there always exists a two-dimensional surface S of minimal area whose boundary coincides with Γ. Beyond studying the properties of these area-minimizing surfaces,such as the presence of branch points or whether they are embedded, there are several natural directions in which to extend the problem by modifying the nature of the prescribed boundary:

- If Γ has more than one connected component, one encounters Plateau-type problems like that of the catenoid in §1.5. In such cases, the surfaces bounded by Γ tend to be topologically more complex.
- By considering higher-dimensional bounding objects, Plateau's Problem can be extended to ask about the existence of volume minimizing objects of arbitrary dimension.
- If the boundary is allowed to move freely within a certain region, one obtains what are known as *free boundary problems*. Courant had already pointed out that his method could be adapted to study such situations.

Continuing from the previous formulation, Plateau's Problem can be extended to higher dimensions using the same tools of differential geometry in Euclidean space as in Chapter 1. In this setting, the role of surfaces is played by submanifolds $M \subseteq \mathbb{R}^n$, defined analogously to regular surfaces. The tangent space and first fundamental form of an m-dimensional submanifold are defined similarly to the 2-dimensional case, but now T_pM is a vector space of dimension m, and the first fundamental form I_p is represented by an $m \times m$ symmetric matrix with coefficients g_{ij}. If (φ, U) is a local parametrization of M, then its volume is given by

$$\mathrm{Vol}(M) := \int_U \sqrt{\det(g)}\ \mathrm{d}^m x.$$

Without going into technicalities, the boundary ∂M of an m-dimensional submanifold will be an $(m-1)$-dimensional submanifold without boundary. We can then state Plateau's Problem in higher dimensions as follows:

Let $m \leq n$ and let $B \subset \mathbb{R}^n$ be an $(m-1)$-dimensional submanifold without boundary. Does there exist an m-dimensional submanifold with boundary M such that $\partial M = B$ and Vol(M) is minimal among all such manifolds?

The integer m is known as the *dimension* of the Plateau Problem and $n - m$ as the *codimension*. This can be further generalized to Riemannian manifolds $(\mathcal{M}, g)$. For a detailed treatment of manifolds and Riemannian geometry, see [Lee12] and [Lee19]; for applications to the theory of minimal surfaces in higher dimensions, see Chapter 1 of [Xin18].

As we mentioned at the end of the previous chapter, the need for more parameters to describe higher-dimensional surfaces creates a challenge for connecting these surfaces with complex analysis. Furthermore, for $m \geq 3$, the existence of isothermal coordinates for an m-dimensional submanifold is not guaranteed, even locally. This makes Douglas's method and similar techniques ineffective when trying to solve Plateau's Problem in higher dimensions. To tackle this, several new theories were developed during the third golden age of minimal surfaces. One of the most notable among these is *Geometric Measure Theory (GMT)*, whose foundations were laid by Herbert Federer and Wendell Fleming.

As the name suggests, GMT approaches Plateau's Problem from a more geometric point of view rather than a purely functional one. Instead of searching for surfaces as images of parameterizations, GMT focuses on minimizing the area of rectifiable currents. To define this concept, one starts with measure theory and introduces the m-dimensional *Hausdorff measure* of a subset $A \subset \mathbb{R}^n$:

$$\mathcal{H}^m(A) := \lim_{\delta \to 0} \inf \left\{ \sum_j \alpha_m \left(\frac{\operatorname{diam}(E_j)}{2} \right)^m : A \subset \bigcup_j E_j,\ \operatorname{diam}(E_j) < \delta \right\},$$

where α_m is the volume of the unit ball in $\mathbb{R}^m$, and $\operatorname{diam}(E_j)$ is the diameter of the set E_j, meaning the greatest distance between any two points in E_j. Intuitively, $\mathcal{H}^m$ measures the "m-dimensional size" of a subset of $\mathbb{R}^n$. For example, if you take a submanifold of dimension 2 inside $\mathbb{R}^n$, its Hausdorff measure will be its area if $m = 2$, but it will be zero if $m > 2$, and infinite if $m < 2$.

A Borel subset $B \subset \mathbb{R}^n$ is called *$\mathcal{H}^m$-rectifiable* if it can be written as a countable union of images of Lipschitz maps and if $\mathcal{H}^m(B) < +\infty$. While rectifiable sets may not be smooth in the usual sense, they still allow a form of differential geometry to be performed on them. Finally, a *rectifiable current* is an oriented rectifiable set with integer multiplicities, finite area, and compact support. The structure of currents allows for the integration of differential forms over them, which means they define linear functionals on differential forms (hence the name *current*). One of the central results in GMT is that the space of rectifiable currents is compact; as a consequence the direct method can be applied to find area minimizing currents. This involves taking a sequence of currents whose areas decrease to the infimum, extracting a convergent subsequence from this sequence, and finally showing that the limit of this subsequence is the desired surface with least area. Thus proving the existence of an area-minimizing current becomes a relatively straightforward task. What remains is to study the regularity of the solution, for instance, whether the area-minimizing current corresponds to a smooth embedded submanifold.

GMT is an extensive and complex theory and a great amount of background knowledge in measure theory, topology, and differential geometry is important. This approach is well explained in Frank Morgan's excellent book [Mor08], which is based on the foundational work of Federer [Fed14]. Over the past decades, GMT has proven successful in dealing with higher-dimensional formulations of Plateau's Problem. This is why it is one of the suggested directions to explore after the classical treatment up to the second golden age of minimal surfaces.

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A Review of complex and harmonic functions

The theory of minimal surfaces intersects many branches of mathematics, making it essential to revisit some foundational material typically covered in an undergraduate curriculum. In the two-dimensional setting, complex analysis proves to be an especially effective tool, and we begin by recalling several classical results; see [BC13] for a comprehensive reference. The second part of this review focuses on harmonic functions in the plane; a detailed exposition can be found in [ABW01].

A.1 Complex analysis

The complex numbers can be constructed in several equivalent ways, for our purposes we will take $\mathbb{C}$ to be the set containing numbers z of the form $a + ib$ where $a, b \in \mathbb{R}$ are known as the real and complex parts respectively and $i^2 = -1$. There is a trivial isomorphism between $\mathbb{C}$ and $\mathbb{R}^2$ given by $a+ib \leftrightarrow (a, b)$ which allows us to apply the notions of euclidean topology and multivariate calculus to the complex numbers, but the fact that we have a well defined multiplication operation between vectors

$$(a + ib)(c + id) = ac - bd + (ad + bc)i$$

will give this space a much richer differentiable structure.

Let $U \subseteq \mathbb{C}$ be open and $f : U \to \mathbb{C}$ be a complex function, we say f is *complex differentiable* at $z_0 \in U$ if the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists in which case we denote it with $f'(z_0)$. If f is complex differentiable at every point in U it is *holomorphic* in U. If $z = x + iy$ and $f = u + iv$ then we can think of each component function as $u = u(x, y)$ and $v = v(x, y)$. The fact that the complex derivative is the best linear approximation of f–which corresponds to multiplying by a complex number–implies that f is holomorphic if it is real differentiable as a function from $\mathbb{R}^2$ to itself and its Jacobian matrix is of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} = r \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

So locally every holomorphic function stretches space by a factor of $r = \sqrt{u_x^2 + u_y^2}$ and rotates it by an angle of $\theta = \arctan\big(u_y/u_x\big)$. In particular, the differential of a holomorphic function will preserve the angles between the original space and the image space as long as $r > 0$–equivalently $f'(z) \neq 0$. Angle preserving functions are known as *conformal* which is why historically holomorphic functions have also been called conformal. This implies that f is holomorphic if and only if it satisfies the *Cauchy-Riemann (CR) equations*

$$u_x = v_y, \quad u_y = -v_x. \tag{A.1.1}$$

In conclusion, holomorphic functions are a particular case of real differentiable functions from $\mathbb{R}^2$ to $\mathbb{R}^2$ with an extra condition on the form of their differential. Although at first it may not seem as a very strong condition, in actuality it is enough to give holomorphic functions an enormous amount of nice properties that are generally not shared with their real counterpart. We remind ourselves that a real analytic function is a function that is infinitely differentiable and admits a power series expansion convergent in an open set, for example $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = (1 + x^2)^{-1}$ is such that

$$f(x) = \sum_{k \geq 0} (-1)^k x^{2k} \quad \text{for } |x| < 1.$$

However there exist functions that are once differentiable but not infinitely differentiable such as $g(x) = |x|\,x$ or functions that are infinitely differentiable but don't admit a series expansions such as $h(x) = \exp\left(x^{-2}\right)$ at $x = 0$. It turns out that holomorphic functions are much more rigid than this.

Theorem A.1.1. *Let $f : U \to \mathbb{C}$ be a complex function, then the following are equivalent:*

(a) f is holomorphic in U;

(b) f is infinitely differentiable in the complex sense;

(c) f is analytic, i.e, it admits a convergent power series in a neighborhood of every point of U.

At first this result is very surprising as we find that requiring a complex function to be once differentiable is enough to find that it is extremely well behaved. Particularly, the behavior of analytic functions is very rigid in the sense that the knowledge of the derivatives of a function at a single point determines the behavior of the function everywhere:

Proposition A.1.2. *Let $f : U \to \mathbb{C}$ be analytic the following are equivalent:*

1. *f is constant;*
2. *All the derivatives of the function vanish at a point $f^{(k)}(z_0) = 0$ for all $k \geq 0$;*
3. *There exist a sequence $\{z_j\}_j$ in U converging to $z_0 \in U$ and a constant $w \in \mathbb{C}$ such that $f(z_j) = w$ for all $j \geq 0$.*

Another important consequence of the complex derivative is found differentiating the Cauchy-Riemann equations

$$\Delta u = u_{xx} + u_{yy} = 0 \quad \text{and} \quad \Delta v = v_{xx} + v_{yy} = 0,$$

functions whose Laplacian vanishes are known as *harmonic*. We also have the converse result:

Theorem A.1.3. *Let u be a harmonic function in a simply connected domain $U \subset \mathbb{R}^2$, then there exists a harmonic function v such that the function*

$$f(z) = u(x, y) + iv(x, y)$$

is holomorphic in U. Moreover, v is uniquely determined up to an additive constant and it is known as the harmonic conjugate of u.

The *conjugation* of complex numbers is the operation that maps $z = x + iy \mapsto \overline{z} = x - iy$. It allows us to define the *modulus* of a complex number $|z| = \sqrt{z\overline{z}}$ which coincides with its real counterpart, namely the euclidean norm of (x, y). We can also retrieve the *real* and *complex* parts of z using the conjugation

$$x = \mathrm{Re}(z) = \frac{z + \overline{z}}{2}, \quad y = \mathrm{Im}(z) = \frac{z - \overline{z}}{2}.$$

From this we can give a formal interpretation of the pair $(z, \overline{z})$ as coordinates of $\mathbb{R}^2$ with the change from (x, y) given by the previous relations. The chain rule allows us to formally define the *Wirtinger derivatives* as

$$\frac{\partial}{\partial z} = \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right), \quad \frac{\partial}{\partial \overline{z}} = \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right)$$

which behave as regular partial derivatives and can be applied to any real valued function. It is an immediate consequence of the Cauchy-Riemann equations that f is holomorphic if and only if $\partial_{\overline{z}} f = 0$ which we can interpret formally as the fact that holomorphic functions do not depend on the complex conjugate $\overline{z}$. A function satisfying $\partial_z f = 0$ is known as a *antiholomorphic* or *anticonformal* map and its conjugate $\overline{f}$ will be holomorphic. A map that is holomorphic and bijective is known as a *biholomorphic* map.

A very special class of complex functions are the *Möbius transformations* which are of the form

$$f(z) = \frac{az + b}{cz + d}$$

for $z \in \mathbb{C}$ and fixed complex coefficients $a, b, c, d \in \mathbb{C}$ such that $ad - bc \neq 0$. These functions are holomorphic everywhere except at the point where the denominator vanishes where they are known as *meromorphic* functions[8]. We can define the *extended complex plane* by adding a point at infinity $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ such that Möbius transforms are bijective and holomorphic in $\mathbb{C}_\infty$ where we understand that $z/\infty = 0$ and $z/0 = \infty$. In the following proposition we take circles in $\mathbb{C}_\infty$ to be either the standard circles in the plane of the form $(x - x_0)^2 + (y - y_0)^2 = r^2$ or straight lines which can be thought of as circles that close at infinity.

Proposition A.1.4. *Let $f : \mathbb{C}_\infty \to \mathbb{C}_\infty$ be a Möbius transform, then:*

(a) f is bijective and meromorphic and its inverse is also a Möbius transform. In particular, the set of Möbius transformations is a group with the composition;

(b) f is determined by fixing the image of three distinct points;

(c) f maps circles in $\mathbb{C}_\infty$ into circles in $\mathbb{C}_\infty$.

A particular family of Möbius transforms with useful properties are the following.

Proposition A.1.5. *Let $a \in \mathbb{C}$ fixed such that $|a| < 1$ and define*

$$f_a(z) = \frac{z - a}{1 - \overline{a}z}, \quad \text{for } z \in \overline{\mathbb{D}}.$$

Then f_a is a $\mathbb{C}$-biholomorphic transformation of the disk mapping $f_a(\mathbb{D}) = \mathbb{D}$ and $f_a(\partial\mathbb{D}) = \partial\mathbb{D}$.

Consider the unit sphere $\mathbb{S}^2$ embedded in $\mathbb{R}^3$ and centered at the origin with the xy plane identified with $\mathbb{C}$. The *stereographic projection* is the map $p : \mathbb{S}^2 \to \mathbb{C}_\infty$ mapping $(a, b, c) \in \mathbb{S}^2$ to the point where the straight line passing trough $(0, 0, 1)$ and x intersect the plane, explicitly

$$(a, b, c) \mapsto \frac{a}{1 - c} + \frac{b}{1 - c}i, \quad (0, 0, 1) \mapsto \infty. \tag{A.1.2}$$

It can be seen that p defines a conformal diffeomorphism between $\mathbb{S}^2$ and $\mathbb{C}_\infty$ which is why the latter is also known as the *Riemann sphere*. In essence, the extended complex plane can be seen as the result of adding a complex structure to the 2-sphere[9]. For example, viewing $\mathbb{C}_\infty$ as a sphere it is clear that straight lines in the plane correspond to circles in the sphere that pass through the north pole. It can also be seen that Möbius transformations on the complex plane correspond to rigid motions of the sphere in $\mathbb{R}^3$. Finally, as a consequence of the classification of isometries of the sphere it can be shown that Möbius transformations are the only biholomorphic mappings from the extended plane to itself.

[8] A meromorphic function is a function that is holomorphic everywhere except at isolated points z_i, known as poles, where the function is of the form $g(z)/h(z)$ for g, h holomorphic and $h(z_i) = 0$.

[9] $\mathbb{C}_\infty$ is also known as the one point compactification of $\mathbb{C}$, which means that by adding a point and an appropriate topology we obtain a compact topological space.

A.2 Harmonic functions

The linear second order partial differential equation

$$\Delta u = 0$$

is known as *Laplace's equation.* $\mathcal{C}^2$ functions having a vanishing Laplacian are known as harmonic and they are highly relevant in many areas of mathematics and physics because of the problems they model and the properties they posses. In this section we will be focusing on harmonic functions in two variables, although most of the theory can be easily generalized to higher dimensions.

One of the most common appearances of harmonic functions is in the context of *boundary problems.* Given an open subset $U \subseteq \mathbb{R}^2$ we look for a function $u : \overline{U} \to \mathbb{R}$ which is harmonic in U and satisfies a prescribed boundary condition, i.e., given a fixed continuous function $g : \partial U \to \mathbb{R}$ we want $u \in \mathcal{C}^2(U) \cap \mathcal{C}^0(\overline{U})$ satisfying

$$\begin{cases} \Delta u = 0 & \text{in } U, \\ u = g & \text{in } \partial U; \end{cases}$$

the question of existence of solutions is known as the *Dirichlet problem* for the Laplace equation. This posing of the problem is what's called a *classical* statement in which we require u to be $\mathcal{C}^2$ and continuous up to boundary but, as we will later see, this is not the only way of posing such problems.

There exist a large number of methods for proving the existence of classical solutions to the Dirichlet problem most of which do not provide explicit formulas for the solution. If the domain of the problem is very simple, having for example a lot of symmetry, then the existence can be proved with the use of Poisson kernels.

Definition A.2.1. The function $P : \mathbb{D} \to [0, +\infty)$ defined in polar coordinates $z = re^{i\theta}$ by

$$P(r, \theta) := \frac{1 - r^2}{1 - 2r\cos(\theta) + r^2},$$

is known as the *Poisson kernel* on the disk.

The Poisson kernel is what's known as a *fundamental solution* to the Laplace equation on the disk since it allows us to construct solutions to given boundary problems via an integral formula.

Proposition A.2.2. *The Poisson kernel has the following properties:*

(a) Limit in the boundary

$$\lim_{r \to 1^-} P(r, \theta) = \begin{cases} 0, & \text{if } \theta \neq 0 \\ \infty, & \text{if } \theta = 0 \end{cases}$$

(b) Normalization

$$\frac{1}{2\pi} \int_0^{2\pi} P(r, \theta - \alpha)\, \mathrm{d}\alpha = 1$$

(c) The Poisson kernel is $\mathcal{C}^\infty(\mathbb{D})$ and is a solution to Laplace's equation for $r < 1$.

Proof. Point a) is a simple computation. For point b) consider the Möbius transformation of the disk given by

$$w = \frac{z - a}{1 - \overline{a}z}, \quad |a| < 1.$$

We know by Prop. A.1.5 that $z = e^{i\alpha} \in \partial\mathbb{D}$ is mapped to $w = e^{i\beta} \in \partial\mathbb{D}$; so substituting, differentiating and taking the modulus gives

$$\frac{\mathrm{d}\beta}{\mathrm{d}\alpha} = \frac{1 - |a|^2}{\left|1 - \overline{a}e^{i\alpha}\right|^2}.$$

Now, changing to polar coordinates we write $a = re^{i\theta}$ for $0 \leq r < 1$ such that

$$\left|\frac{\mathrm{d}\beta}{\mathrm{d}\alpha}\right| = \frac{1 - r^2}{1 - 2r\cos(\theta - \alpha) + r^2}$$

which is precisely $P(r, \theta - \alpha)$. Using this fact we compute via a change of variables

$$\int_0^{2\pi} P(r, \theta - \alpha)\ \mathrm{d}\alpha = \int_0^{2\pi} \mathrm{d}\beta = 2\pi.$$

The infinite differentiability of P is clear from its definition and to verify it is harmonic we use the form of the Laplacian in polar coordinates

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{1}{r}\frac{\partial}{\partial r} + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2}$$

to see that $\Delta P = 0$ which proves point c). $\square$

Properties (a) and (b) tell us that $(2\pi)^{-1}P$ behaves like a Dirac Delta function at boundary of the disk and property (c) says that it is harmonic. We can now state and prove the main result involving Poisson's kernel.

Theorem A.2.3 (Poisson's Formula)**.** *Let $g : \partial\mathbb{D} \to \mathbb{R}$ fixed and continuous. The function $u : \overline{\mathbb{D}} \to \mathbb{R}$ defined by*

$$u(re^{i\theta}) = \begin{cases} \frac{1}{2\pi}\int_0^{2\pi} P(r, \theta - \tau) g(e^{i\tau})\ \mathrm{d}\tau, & \text{if } r < 1 \\ g(e^{i\theta}), & \text{if } r = 1 \end{cases}$$

is a classical solution to the Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } \mathbb{D}, \\ u = g & \text{in } \partial\mathbb{D}; \end{cases}$$

with $u \in \mathcal{C}^\infty(\mathbb{D}) \cup \mathcal{C}^0(\overline{\mathbb{D}})$.

Proof. The fact that u is harmonic and $\mathcal{C}^\infty$ in $\mathbb{D}$ is because P is

$$\Delta u(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} \Delta P(r, \theta - \tau) g(e^{i\tau})\ \mathrm{d}\tau = 0.$$

Now, we just have to show that u is continuous up to the boundary, that is for each θ we have

$$\lim_{r \to 1^-} u(re^{i\theta}) = g(e^{i\theta}).$$

Indeed, from the normalization property of P we have that

$$\left|u(re^{i\theta}) - g(e^{i\theta})\right| = \left|\frac{1}{2\pi}\int_0^{2\pi} P(r, \theta - \tau)\left[g(e^{i\tau}) - g(e^{i\theta})\right]\ \mathrm{d}\tau\right| \leq \frac{1}{2\pi}\int_0^{2\pi} P(r, \theta - \tau)\left|g(e^{i\tau}) - g(e^{i\theta})\right|\ \mathrm{d}\tau.$$

Let $\epsilon > 0$ and $\delta > 0$ such that $|\tau - \theta| < \delta$ implies $\left|g(e^{i\tau}) - g(e^{i\theta})\right| < \epsilon/2$ which exists by continuity of g

and the complex exponential. We can then split the previous integral so that

$$\begin{aligned}\frac{1}{2\pi}\int_0^{2\pi} P(r,\theta-\tau)\left|g(e^{i\tau})-g(e^{i\theta})\right| \ \mathrm{d}\tau = \frac{1}{2\pi}\int_{|\tau-\theta|<\delta} P(r,\theta-\tau)\left|g(e^{i\tau})-g(e^{i\theta})\right| \ \mathrm{d}\tau \\ + \frac{1}{2\pi}\int_{|\tau-\theta|\geq\delta} P(r,\theta-\tau)\left|g(e^{i\tau})-g(e^{i\theta})\right| \ \mathrm{d}\tau.\end{aligned}$$

For the first integral we have

$$\frac{1}{2\pi}\int_{|\tau-\theta|<\delta} P(r,\theta-\tau)\left|g(e^{i\tau})-g(e^{i\theta})\right| \ \mathrm{d}\tau \leq \frac{\epsilon}{4\pi}\int_0^{2\pi} P(r,\theta-\tau)\ \mathrm{d}\tau = \frac{\epsilon}{2},$$

and for the second let $M>0$ such that $\left|g(e^{i\theta})\right| \leq M$ for any $\theta\in\mathbb{R}$ and take r close enough to 1 such that

$$P(r,\theta-\tau) < \frac{\epsilon}{4M}$$

which is possible since $|\tau-\theta|\geq\delta$ and P goes to 0 for non zero angles so

$$\frac{1}{2\pi}\int_{|\tau-\theta|\geq\delta} P(r,\theta-\tau)\left|g(e^{i\tau})-g(e^{i\theta})\right| \ \mathrm{d}\tau \leq \frac{2M}{2\pi}\int_0^{2\pi} \frac{\epsilon}{4M}\ \mathrm{d}\tau = \frac{\epsilon}{2}.$$

All in all, for r close enough to 1 we will have

$$\left|u(re^{i\theta})-g(e^{i\theta})\right| \leq \epsilon,$$

which proves our claim. □

We note that the unit disk is not special in any particular way since the Laplace equation is translation and scaling invariant, i.e., if u is harmonic then so are $u(x+a)$ and $u(\lambda x)$ for $a\in\mathbb{R}^2$ and $\lambda\in\mathbb{R}$. This implies that the previous Poisson kernel can be modified to work in any ball $B\subset\mathbb{R}^2$ and any properties we deduce for $\mathbb{D}$ will also hold in B. For example, if $u:\overline{\mathbb{D}}\to\mathbb{R}$ is already a solution to a Dirichlet problem then

$$u(re^{i\theta}) = \frac{1}{2\pi}\int_0^{2\pi} P(r,\theta-\tau)u(e^{i\tau})\ \mathrm{d}\tau$$

holds trivially and we can compute the average value of the function on any circle $|z|=r<1$ by integrating and exchanging the order of integration

$$\begin{aligned}\frac{1}{2\pi}\int_0^{2\pi} u(re^{i\theta})\ \mathrm{d}\theta &= \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{1}{2\pi}\int_0^{2\pi} P(r,\theta-\tau)u(e^{i\tau})\ \mathrm{d}\tau\right]\ \mathrm{d}\theta \\ &= \frac{1}{2\pi}\int_0^{2\pi}\left[\frac{1}{2\pi}\int_0^{2\pi} P(r,\theta-\tau)\ \mathrm{d}\theta\right]u(e^{i\tau})\ \mathrm{d}\tau = \frac{1}{2\pi}\int_0^{2\pi} u(e^{i\tau})\ \mathrm{d}\tau = u(0)\end{aligned}$$

where we have used that $P(0,\theta-\tau)=1$ in the last equality. We have found that the average value of the function in any circle centered at the origin is precisely the value at the origin. Because of the previous observation this result holds true more generally.

Proposition A.2.4 (Mean Value Property). *Let $U\subseteq\mathbb{R}^2$ open and $a\in U$, $r>0$ such that $B(a,r)\subset U$. If $u:U\to\mathbb{R}$ is a harmonic function then*

$$u(a) = \frac{1}{2\pi}\int_0^{2\pi} u(a+re^{i\theta})\ \mathrm{d}\theta.$$

This result has the following important consequence.

Theorem A.2.5 (Maximum Principle). *Let $u : U \to \mathbb{R}$ a harmonic function and suppose that $a \in U$ is a local maximum of u, then u is constant.*

Proof. We have that for r small enough $u(z) \leq u(a)$ for $|z-a| \leq r$ and

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + re^{i\theta}) \, \mathrm{d}\theta$$

since the right hand side is the average it must be that $u(a + re^{i\theta}) = u(a)$ for all θ. Applying this argument for any r we find that u is constant in an open disk. Let v be the harmonic conjugate of u in U which exists by Theorem A.1.3 and define the complex function $f = u + iv$. Since u is constant in an open disk so will v by the CR equations which implies f will be constant in the disk. Using the rigidity of analytic functions (Prop. A.1.2) we find f will constant in U and so will u. □

As a consequence of the previous proof we also find the following result.

Corollary A.2.6. *Harmonic functions are real analytic.*

It is clear that there is an analogous Minimum Principle which can be derived by considering $-u$ in the result. In particular, this implies that a harmonic function has to attain its extreme values in the boundary of its domain.

Corollary A.2.7. *Let $u : \overline{U} \to \mathbb{R}$ continuous and harmonic in U, then*

$$\inf_{\partial U} u = \inf_{U} u \quad \textit{and} \quad \sup_{\partial U} u = \sup_{U} u.$$

Using this we come full circle to prove the uniqueness of the solutions to the Dirichlet problem on the disk.

Theorem A.2.8. *There exist a unique solution to a Dirichlet problem for the Laplacian on the disk.*

Proof. We have already shown the existence with Poisson's kernel. For the uniqueness suppose u and v are two solutions to the same Dirichlet problem and define $w = u - v$ which is also harmonic by the linearity of the Laplacian. Using the previous result we find that $\sup_{\mathbb{D}} w = 0$ since $u = v$ on the boundary. Similarly we find that $\inf_{\mathbb{D}} w = 0$ so $w = 0$ implying $u = v$ in $\overline{\mathbb{D}}$. □

Another useful consequence of the Maximum Principle is that the harmonic property is preserved under uniform convergence.

Proposition A.2.9. *Let $u_k : \overline{\mathbb{D}} \to \mathbb{R}$ be a sequence of continuous and harmonic functions in $\mathbb{D}$. If u_k converges uniformly then the limit is also harmonic in the disk.*

Proof. Define the sequence $g_k : \partial\mathbb{D} \to \mathbb{R}$ by $g_k = u_k|_{\partial\mathbb{D}}$ which converges uniformly since u_k does. Set g the limit of g_k and let $u : \overline{\mathbb{D}} \to \mathbb{R}$ be the unique solution to the Dirichlet problem of g. Now, let $\epsilon > 0$ such that $\big|g_k(z) - g(z)\big| < \epsilon$ for k large enough, then

$$\max_{z \in \overline{\mathbb{D}}} u_k(z) - u(z) = \max_{z \in \partial\mathbb{D}} u_k(z) - u(z) = \max_{z \in \partial\mathbb{D}} g_k(z) - g(z) < \epsilon,$$

and the same holds true for the minimum of the difference so we find $u_k \to u$ uniformly on $\overline{\mathbb{D}}$ and we finish by noting that u is harmonic. □

We end this section with a result we will need in the proof of Plateau's problem.

Lemma A.2.10. *Let $u_k : \overline{\mathbb{D}} \to \mathbb{R}$ be a uniformly converging sequence of harmonic functions in the disk with limit u. Then ∇u_k converges uniformly to ∇u on any proper compact $K \subset \mathbb{D}$.*

Proof. WLOG we can assume $K = B(0, R)$ for some $R < 1$. Using the Poisson formula

$$\left|\nabla u_k(re^{i\theta}) - \nabla u(re^{i\theta})\right| \leq \frac{1}{2\pi}\int_0^{2\pi} \left|\nabla P(r, \theta - \tau)\right|\left|u_k(e^{i\tau}) - u(e^{i\tau})\right| \, \mathrm{d}\tau$$

and conclude using that $u_k \to u$ uniformly on $\partial\mathbb{D}$ and $|\nabla P|$ is bounded for $r \leq R$. □

B Integration theory results

We remind ourselves that a sufficient condition for the Riemann integral of a function to be well defined is that it is continuous and integrated over a compact set. Both $A(\varphi)$ and $E(\varphi)$ in Chapter 2 are defined as integrals of positive functions over $\mathbb{D}$ so, even if we do not know whether the integrand can be extended continuously to the boundary, whenever the integrals do not take finite values it makes sense to say they evaluate to $+\infty$.

Throughout the thesis we use frequently the following two results which can be found in any standard multivariate calculus book or course.

Theorem B.0.1 (Integration by parts)**.** *Let $D \subseteq \mathbb{R}^n$ be a bounded domain with a $\mathcal{C}^1$ boundary ∂D, i.e. which can be written locally as the graph of a $\mathcal{C}^1$ function. Then for any $f, g \in \mathcal{C}^1(\overline{D})$ we have*

$$\int_D \partial_i f \, g \, \mathrm{d}^n x = -\int_D f \, \partial_i g \, \mathrm{d}^n x + \int_{\partial D} f \, g \, \nu_i \, \mathrm{d}S$$

where v is the unit normal vector to ∂D and $i = 1, \ldots, n$.

Theorem B.0.2 (Differentiation under the integral sign)**.** *Let $D \times [a, b] \subseteq \mathbb{R}^n \times \mathbb{R}$, with D compact and $f : D \times [a, b] \to \mathbb{R}$ such that:*

(a) for each $t_0 \in [a, b]$ fixed the function $f(x_1, \ldots, x_n; t_0)$ is Riemann integrable on D,

(b) for every $(x_1, \ldots, x_n; t) \in D \times [a, b]$, f is continuously differentiable with respect to t and $\partial_t f$ is Riemann integrable on D;

then

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_D f(x_1, \ldots, x_n; t) \, \mathrm{d}^n x = \int_D \frac{\partial}{\partial t} f(x_1, \ldots, x_n; t) \, \mathrm{d}^n x.$$

Finally, we also state and prove the continuous version of the Fundamental Lemma of the Calculus of Variations.

Theorem B.0.3 (Fundamental Lemma of the Calculus of Variations)**.** *Let $D \subseteq \mathbb{R}^n$ a domain and $f : D \to \mathbb{R}$ be a continuous function such that*

$$\int_D f \, \eta \, \mathrm{d}^n x = 0$$

for all $\eta \in \mathcal{C}_c^\infty(D)$, then $f = 0$ identically in D.

Proof. WLOG suppose there exists balls $B \subset B' \subset D$ such that $f(x) > m > 0$ for all $x \in B'$. Choose $\eta \in \mathcal{C}_c^\infty(B') \subset \mathcal{C}_c^\infty(D)$ such that $\eta \geq 0$ and $\eta|_B = 1$, then

$$\int_D f \, \eta \, \mathrm{d}x = \int_{B'} f \, \eta \, \mathrm{d}x \geq \int_B f \, \eta \, \mathrm{d}x = m\mathrm{Vol}(B) > 0$$

where $\mathrm{Vol}(B)$ is the volume of the ball. We have arrived at a contradiction so $f = 0$ everywhere. □

C Even more on isothermal parameters

We take a look at the additional assumptions made on the admissible parametrizations in X_Γ such that the existence of isothermal parameters is ensured.

We begin by deriving the condition in Eq. (2.2.2). Assuming that φ is an immersion, we want $|\mu| \leq k < 1$, that is

$$\frac{(E+G-2\sqrt{EG-F^2})^2}{(E-G)^2+4F^2} \leq 1-\delta$$

for some $\delta > 0$. This is equivalent to

$$\begin{aligned}
&E+G-\sqrt{1-\delta}\sqrt{(E-G)^2+4F^2} \leq 2\sqrt{EG-F^2} \\
&\iff (E+G)^2+(1-\delta)((E-G)^2+4F^2)-2(E+G)\sqrt{1-\delta}\sqrt{(E-G)^2+4F^2} \leq 4(EG-F^2) \\
&\iff (1-\delta/2)\sqrt{(E-G)^2+4F^2} \leq (E+G)\sqrt{1-\delta} \\
&\iff (1-\delta+\delta^2/4)((E-G)^2+4F^2) \leq (E^2+2EG+G^2)(1-\delta) \\
&\iff \frac{\delta^2}{16(1-\delta)} \leq \frac{EG-F^2}{(E-G)^2+4F^2},
\end{aligned}$$

and given that $EG \geq F^2$ we have that

$$\frac{\delta^2}{16(1-\delta)} \leq \frac{EG-F^2}{(E+G)^2} \iff \frac{\delta}{4\sqrt{1-\delta}} \leq \frac{\sqrt{EG-F^2}}{E+G}$$

is a sufficient condition and it is precisely the condition in Eq. (2.2.2). This has the immediate consequence that $\mathcal{A}_\Gamma < \infty \implies \mathcal{E}_\Gamma < \infty$. Now, if φ is not an immersion but we take $\varphi_\epsilon = (\varphi, \epsilon u, \epsilon v)$ then we have seen that this is an immersion and has Beltrami coefficient

$$\mu_\epsilon = \frac{E+G+2\epsilon^2-2\sqrt{EG-F^2+\epsilon^2(E+G)+\epsilon^4}}{E-G-2iF}$$

which satisfies $|\mu_\epsilon| \leq k < 1$ under the sufficient condition

$$\inf_{(u,v)\in\mathbb{D}} \frac{\sqrt{EG-F^2+\epsilon^2(E+G)+\epsilon^4}}{E+G} > 0$$

and this is satisfied under condition (2.2.2) by the φ's that need not be immersions.

As for the Hölder continuity, it was proved by Lichtenstein in 1916 that global isothermal parameters always exist provided the bounding curve is $\mathcal{C}^{1,\alpha}$ and so is the parametrization. This of course can also be applied to the Plateau Problem that is solved in Chapter 2 but then we have to require even more regularity on Γ. See [HM05] for an overview of the result, but bare in mind that the proof of the article is not the original one and if applied to our case would constitute a circular argument since it relies on the solution to Plateau's Problem. The takeaway is that Hölder continuity is expected for isothermal parameters to exist.